

Joint Estimation Of State and Parameters for a Class of Nonlinear Disturbed Systems with Unknown Noise

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Abstract

This paper presents a Physics-Informed Bank of Estimators (PIBE) framework for jointly estimating states, parameters, and disturbances for a class of nonlinear disturbed systems in lower-triangular form. First, we introduce a bank-of-estimators approach for recursive state, parameter, and disturbance estimation, where each cell uses upstream estimates and a global objective backpropagates through the preceding cells during joint fine-tuning. Subsequently, we extend the method to compactly supported measurement-noise laws, including truncated-Gaussian and non-Gaussian cases. For the non-Gaussian case, we combine Physics-Informed Neural Networks (PINNs) with constrained scalar Energy-Based Models (EBMs) to fit flexible pooled residual densities. We later address error propagation through cascaded approximators and derive a conditional upper bound on the estimation error. The effectiveness of the proposed method is demonstrated through extensive numerical simulations.

Key words: State and parameter estimation, Physics-informed neural networks, Nonlinear systems, Energy-based models, Non-Gaussian noise, Observer design

1 Introduction

Physical systems are commonly represented by nonlinear dynamical models that are subject to model disturbances [19]. In a wide range of applications, accessing the system's state is crucial for control design, monitoring, fault detection, and diagnosis (see [14,28,29] for more details). However, in practice, measuring states directly is not always feasible due to physical or technological constraints. When direct measurements are unavailable, estimation algorithms, referred to as observers, are developed. The control community has extensively studied this problem, resulting in a wide variety of observers tailored to different system structures and available measurements (see [3] for a comprehensive literature review). Examples include Kalman-like filters (KF, EKF, UKF) [13,12], sliding mode observers (SMO) [32],

high-gain observers for systems in triangular canonical form [18,9], and Kazantzis–Kravaris/Luenberger (KKL) observers [16,2] based on a nonlinear change of coordinates that conjugates the dynamics to a stable linear form. While observer and estimation design for linear systems is well established, developing observers for nonlinear systems remains challenging. Moreover, a key challenge for these model-based observers is their reliance on an accurate dynamic model, which can lead to reduced performance in the presence of uncertainties; therefore, one needs to design observers robust to uncertainties. In addition to the system state, many physical systems depend on unknown parameters that may be constant or time-varying. Therefore, classical observers have been extended into adaptive observers [4] that jointly estimate states and parameters.

Beyond model uncertainty and unknown parameters, a second challenge in state and parameter estimation arises from the nature of the available measurements. In real-world scenarios, measurements are often noisy, and many proposed algorithms assume Gaussian noise profiles. However, this assumption can be too simplis-

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tic in many domains, such as bioengineering, geology, and economics, where non-Gaussian noise is prevalent. To address the limitations highlighted above, learning-based estimation algorithms have recently emerged, providing new approaches to state estimation through Neural Networks (NNs) that complement classical observer and parameter estimation techniques. Two main directions have been explored: a learning algorithm can itself serve as an estimator, or it can be used as a solver for a standard observer. For example, in [23], the authors propose learning the mapping of the nonlinear Kazantzis–Kravaris/Luenberger observer, an approach later extended to time-varying output-delay systems in [22]. Similarly, a Recurrent Neural Network (RNN)-based nonlinear observer that combines RNNs with the Extended Luenberger Observer (ELO) to effectively capture nonlinear system dynamics is introduced in [1]. These studies, among others, have paved the way for learning-based observer design, enabling new strategies for solving the observation and estimation problem. However, treating learning algorithms as estimators introduces two important limitations: the need for large amounts of training data and the lack of formal convergence guarantees, which motivate the design of estimators that exploit structural knowledge to reduce data requirements and provide quantitative error bounds.

In this paper, we propose a learning-based method for jointly estimating the state, parameters, and disturbance of nonlinear dynamical systems in a specific observability canonical form. Leveraging the Physics-Informed Neural Network (PINN) framework, we embed the system’s ordinary differential equation (ODE) residual into the training loss, thereby reducing data requirements while exploiting the structural information already used by classical observers. The designed chain of estimators then achieves accurate state and parameter estimation for disturbed nonlinear systems while simultaneously estimating unknown disturbances. Inspired by [27], we also integrate PINNs with constrained scalar EBM so that compactly supported non-Gaussian residual densities can be fitted without prescribing a named parametric family. This flexibility still requires explicit support and regularity assumptions and does not, by itself, guarantee robustness or identifiability. A preliminary version of this work was presented at the 2026 European Control Conference [5], where the PIBE framework was introduced with an error-propagation bound limited to state estimation under absent or Gaussian measurement noise. The present paper extends that version in three directions: it introduces the Energy-Based PIBE (EPIBE) for non-Gaussian measurement-noise scenarios, generalizes the error analysis to the joint estimation of states, parameters, and disturbance, and validates the framework on a higher-order nonlinear system including non-Gaussian scenarios. The main contributions of this work are: (i) we develop a PINN-based bank of estimators (PIBE) capable of jointly estimating states, parameters, and disturbance

for nonlinear systems in observability canonical form; (ii) we extend it to an Energy-Based PINN framework (EPIBE) that jointly fits a constrained pooled residual-density model and the physical estimator; (iii) we derive a conditional upper bound on error propagation within the chain under the stated regularity, Lipschitz, and post-training approximation assumptions. The paper is organized as follows: Section 2 describes the class of nonlinear systems considered. Section 3 details the proposed framework for the joint estimation of states, parameters, and disturbances under compactly supported truncated-Gaussian and non-Gaussian noise. Section 4 analyses the propagation of estimation error and derives a bound on the cumulated error. Section 5 presents numerical results, and Section 6 concludes the paper.

2 Preliminaries and problem formulation

We consider the following nonlinear system with autonomous drift, an additive time-varying disturbance in the last state equation, and measurement noise, written in triangular canonical form with multiple nonlinearities:

$$\begin{cases} \dot{x} = \begin{bmatrix} \dot{x}_1 \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} x_2 + f_1(x_1, \theta_1) \\ \vdots \\ x_n + f_{n-1}(x_1, \dots, x_{n-1}, \theta_1, \dots, \theta_{n-1}) \\ f_n(x, \theta) + d(t) \end{bmatrix}, \\ y = x_1 + \omega(t), \end{cases} \quad (1)$$

where $n \geq 2$ denotes the state dimension, $x = [x_1, \dots, x_n]^\top$ is the system’s state vector with $x(t) \in \mathbb{R}^n$, and $y(t) \in \mathbb{R}$ is the measured output signal. For $i = 1, \dots, n-1$, the known nonlinearity f_i has the signature $f_i(x_1, \dots, x_i, \theta_1, \dots, \theta_i)$, whereas the last known nonlinearity is $f_n(x, \theta)$. The unknown constant parameter vector is $\theta = (\theta_1, \dots, \theta_{n-1}) \in \Theta \subset \mathbb{R}^{n-1}$, where

$$\Theta = \Theta_1 \times \dots \times \Theta_{n-1}$$

and every $\Theta_j \subset \mathbb{R}$ is nonempty and compact. On the finite estimation horizon $[0, T]$, the disturbance is assumed to admit the decomposition

$$d(t) = \Gamma_q(t)^\top a + r_q(t), \quad (2)$$

where

$$\Gamma_q(t) = [\gamma_0(t) \ \dots \ \gamma_{q-1}(t)]^\top$$

is a known vector of basis functions $\gamma_i \in C^1([0, T])$, $i = 0, \dots, q-1$, $a \in \mathcal{A} \subset \mathbb{R}^q$ is an unknown constant coefficient vector, and $\mathcal{A}$ is compact. The basis functions are assumed to be linearly independent on $[0, T]$, equivalently,

$$W_\Gamma := \int_0^T \Gamma_q(t) \Gamma_q(t)^\top dt \succeq \underline{\gamma}_d I_q, \quad \underline{\gamma}_d > 0. \quad (3)$$

The term $r_q \in L^\infty([0, T])$ accounts for the disturbance component not represented by the selected basis and

satisfies

$$\|r_q\|_{L^\infty(0,T)} \leq \varepsilon_{d,q}. \quad (4)$$

The exact finite-dimensional disturbance class is recovered when $\varepsilon_{d,q} = 0$. Since $\mathcal{A}$ is compact and Γ_q is continuous on $[0, T]$, the structured component $\Gamma_q^\top a$ is uniformly bounded; hence $d \in L^\infty([0, T])$. Finally, ω denotes a measurement-noise process whose admissible realizations satisfy $\|\omega\|_{L^\infty(0,T)} \leq \bar{w}$. Whenever densities, moments, or expectations are used, ω is understood as a random process on an underlying probability space with compactly supported one-dimensional laws. Accordingly, the Gaussian case considered below refers to a zero-mean Gaussian law symmetrically truncated to a fixed compact interval. No independence between time samples is assumed unless it is stated explicitly.

Remark 1 *The triangular structure in (1) is the observability canonical form underlying the design of high-gain and cascade observers [9, 18]. This structure is restrictive, but it represents a broad subclass of single-output nonlinear systems that can be mapped locally into an observability normal form under the appropriate observability-rank and coordinate-regularity conditions [17].*

Assumption 1 (Finite-horizon well-posedness and compact admissibility)

The sets $\chi_0 \subseteq \chi \subset \mathbb{R}^n$ are compact. For every $x_0 \in \chi_0$, $\theta \in \Theta$, $a \in \mathcal{A}$, and every remainder r_q satisfying (4), System (1) has a unique solution $x(\cdot; x_0, \theta, a, r_q)$ on $[0, T]$, and this solution remains in χ . For every admissible noise realization, the corresponding measured output remains in a fixed compact set $\mathcal{Y} \subset \mathbb{R}$ on $[0, T]$.

System (1) can be written in the following compact form:

$$\begin{cases} \dot{x}(t) = f(x(t), \theta) + E_d d(t), \\ y(t) = Cx(t) + \omega(t), \end{cases} \quad (5)$$

where $E_d = [0, \dots, 0, 1]^\top$ and $C = [1, 0, \dots, 0]$. We assume that $f : \chi \times \Theta \rightarrow \mathbb{R}^n$ admits a C^1 extension to an open neighborhood of the compact set $\chi \times \Theta$. In particular, f is locally Lipschitz there, and the partial derivatives used in the sensitivity analysis of Section 4 exist and are bounded on the admissible set. Having the system's output measurements and the known nonlinearities f_i , this study aims at estimating the states $x_k(t)$ for $k = 2, \dots, n$, together with the parameters $\theta = (\theta_1, \dots, \theta_{n-1})$ and the time-varying unknown disturbance $d(t)$. For this purpose, we propose a bank of PINN-based estimators for recursive estimation of states, parameters, and disturbance. In this context, a bank of estimators refers to a sequence of estimators cascaded together, each one taking as input the concatenated outputs of the previous ones.

2.1 Physics Informed Neural Networks

Physics-Informed Neural Networks are a class of NNs that incorporate prior knowledge about a physical

model, typically described by Ordinary Differential Equations (ODEs) or Partial Differential Equations (PDEs) [30]. By embedding physical principles directly into the network's architecture, PINNs enforce these constraints, reducing the reliance on large datasets and enhancing the model's generalization capabilities [15]. The classical formulation of PINNs, as introduced by [30], involves a multi-layer perceptron (MLP) designed to approximate the solution $x_\phi(t)$ of a dynamical model, where $x \in \Phi \subset \mathbb{R}^m$ and $t > 0$ and ϕ represent the optimized parameters. The physical model is encoded into the loss function, creating a hybrid data-physics loss that takes the following form:

$$\mathcal{L} = \mathcal{L}_{Data} + \lambda \mathcal{L}_{Physics}, \quad (6)$$

where $\mathcal{L}_{Data}$ represents the error between the neural network's prediction and the observed data, and $\mathcal{L}_{Physics}$ is the term that enforces the satisfaction of the physical laws (e.g., the ODE residuals), $\lambda > 0$ being a weighting parameter. In the following, $\mathcal{L}_{Physics}$ refers to the square sum of residuals (SSR). Unsupervised learning with PINNs is also possible when data is unavailable; in such cases, the loss function is derived entirely from the governing physical laws, as discussed in [6].

2.2 Energy-Based Models

Energy-Based Models (EBMs), originally proposed in [20], constitute a versatile class of probabilistic models for learning data-driven distributions [20]. They have demonstrated their effectiveness in numerous domains, such as visual data modeling and generation [7], [25], regression-based inference, and noise modeling tasks [27]. The core idea of EBMs is to associate each generic inputoutput pair (u, v) with a scalar energy function $\hat{E}_\zeta(u, v)$, parameterized by a neural network with parameters ζ . Given this energy function, the conditional probability of v given u is defined as:

$$p(v|u, \hat{E}_\zeta) = \frac{e^{-\beta \hat{E}_\zeta(u, v)}}{Z(u, \hat{E}_\zeta)}, \quad (7)$$

$$Z(u, \hat{E}_\zeta) = \int e^{-\beta \hat{E}_\zeta(u, \tilde{v})} d\tilde{v}, \quad (8)$$

where β is an inverse temperature parameter, typically set to 1. The normalization constant $Z(u, \hat{E}_\zeta)$, also referred to as the partition function, ensures that $p(v|u, \hat{E}_\zeta)$ defines a valid probability density. Despite their flexibility, training EBMs is computationally demanding because $Z(u, \hat{E}_\zeta)$ is usually intractable, especially when v lies in a high-dimensional space. Consequently, numerical techniques like Monte Carlo (MC) estimation are often required to approximate this integral. In practice, given a training set of samples $\{(u_i, v_i)\}_{i=1}^N$, the objective is to minimize the negative

log-likelihood (NLL), which can be expressed as:

$$\begin{aligned} \text{NLL}(\{u, v\}, \hat{E}_\zeta) &= -\log \left(\prod_{i=1}^N p(v_i | u_i, \hat{E}_\zeta) \right) \\ &= \sum_{i=1}^N \left[\log Z(u_i, \hat{E}_\zeta) + \beta \hat{E}_\zeta(u_i, v_i) \right]. \end{aligned} \quad (9)$$

This formulation highlights the key training challenge of EBMs: estimating or efficiently approximating the partition function during optimization. In the present work, the response variable is a scalar consistency residual and the energy is not conditioned on an external input. We therefore write $E_\zeta(\xi)$, identify $\zeta \equiv \Theta_{EBM}$, and use the compact, gauge-normalized density class specified later in Assumption 3.

3 PIBE-based states, parameters, and disturbance estimation

In this section, we propose a PINN-based approach to jointly estimate the state, parameters, and disturbance of the system described by Equation (1) from the measured output $y(t)$.

Assumption 2 (*Joint finite-horizon identifiability and local conditioning*) Let $y_0(t) := Cx(t)$ denote the noise-free output and define the joint unknown vector and its admissible set by

$$\eta := (x_0, \theta, a), \quad \mathcal{P} := \chi_0 \times \Theta \times \mathcal{A}.$$

For the nominal finite-dimensional disturbance model $d(t) = \Gamma_q(t)^\top a$, corresponding to $r_q = 0$ in (2), define the finite-horizon output map

$$\begin{aligned} \mathcal{O}_T : \mathcal{P} &\longrightarrow L^2(0, T), \\ \eta &\longmapsto y_0(\cdot; \eta). \end{aligned} \quad (10)$$

The following properties hold:

(i) Structural identifiability: for every $\eta, \tilde{\eta} \in \mathcal{P}$,
 $\mathcal{O}_T(\eta) = \mathcal{O}_T(\tilde{\eta}) \quad \text{a.e. on } [0, T] \implies \eta = \tilde{\eta}. \quad (11)$

(ii) Quantitative local conditioning: for the true vector $\eta^* = (x_0^*, \theta^*, a^*)$, there exist a neighborhood $\mathcal{N}(\eta^*)$ and a constant $c_{\text{id}} > 0$ such that

$$\|\mathcal{O}_T(\eta) - \mathcal{O}_T(\eta^*)\|_{L^2(0, T)} \geq c_{\text{id}} \|\eta - \eta^*\| \quad (12)$$

for every $\eta \in \mathcal{N}(\eta^*) \cap \mathcal{P}$. When $r_q \neq 0$, it is treated as the bounded model mismatch specified in (4); consequently, exact recovery is not asserted and the resulting bounds retain a term depending on $\varepsilon_{d,q}$.

3.1 Proposed Physics-Informed Bank of Estimators framework

In this subsection, we present the PIBE framework for the truncated-Gaussian case, which also covers the noise-

free setting. If the first-cell noise marginals are zero-mean Gaussian laws with a common variance, symmetrically truncated to a fixed admissible support, then, provided that the candidate residuals remain in that support, the first-cell mean-squared-error term coincides, up to a positive scale and an additive constant, with the sum of the marginal negative log-likelihoods. Without temporal independence this sum is a composite likelihood, not the joint trajectory likelihood. The downstream quadratic terms are deterministic inter-cell consistency penalties; no Gaussian propagation through the bank is assumed. The estimator is conditioned on the measured trajectory and on the upstream estimates, as described next.

For each state coordinate, let $\mathcal{X}_j \subset \mathbb{R}$ be a fixed compact interval containing the j -th coordinate projection of χ . All state decoders and parameter heads are constrained to the corresponding admissible sets $\mathcal{X}_j$ and Θ_j ; bounded output reparameterizations may be used when these sets are intervals. This convention supplies the compact output ranges used below and ensures, recursively, that the finite-dimensional trajectory-input sets are compact.

Let $\ell \in \{1, \dots, P\}$ index a trajectory and let $\{t_i\}_{i=1}^N \subset [0, T]$ be the fixed data grid. Define the sampled measurement vector

$$\mathbf{Y}^\ell := [y^\ell(t_1) \cdots y^\ell(t_N)]^\top \in \mathbb{R}^N, \quad (13)$$

and, for each upstream state estimate,

$$\hat{\mathbf{X}}_j^{j,\ell} := [\hat{x}_j^{j,\ell}(t_1) \cdots \hat{x}_j^{j,\ell}(t_N)]^\top. \quad (14)$$

For $k = 2, \dots, n+1$, the trajectory-level input to the k -th cell is

$$\begin{aligned} \mathbf{U}_{k-1}^\ell &:= \text{col}(\mathbf{Y}^\ell, \hat{\mathbf{X}}_2^{2,\ell}, \dots, \hat{\mathbf{X}}_{k-1}^{k-1,\ell}, \\ &\quad \hat{\theta}_1^{2,\ell}, \dots, \hat{\theta}_{k-2}^{k-1,\ell}) \in \mathcal{U}_{k-1}, \end{aligned} \quad (15)$$

where empty sequences are omitted, so that $\mathbf{U}_1^\ell = \mathbf{Y}^\ell$. More precisely, with empty products interpreted as singletons, the admissible input set is

$$\begin{aligned} \mathcal{U}_{k-1} &:= \mathcal{Y}^N \times \prod_{j=2}^{k-1} \mathcal{X}_j^N \\ &\quad \times \prod_{j=1}^{k-2} \Theta_j \subset \mathbb{R}^{d_{k-1}}, \\ d_{k-1} &= N(k-1) + (k-2). \end{aligned} \quad (16)$$

which is compact.

For every $k = 2, \dots, n$, the k -th PIBE cell consists of a trajectory encoder, a continuous-time state decoder, and a time-independent parameter head. First, the encoder computes

$$\mathbf{z}_{k-1}^\ell = \mathcal{E}_k(\mathbf{U}_{k-1}^\ell; \Theta_{E_k}) \in \mathbb{R}^{r_k}. \quad (17)$$

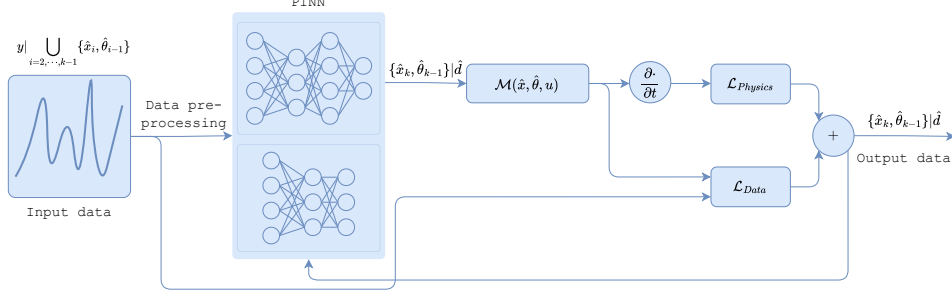


Fig. 1. A unitary estimation cell of the proposed Physics-Informed Bank of Estimators (PIBE) estimator.

The state decoder is the map

$$\mathcal{S}_k : [0, T] \times \mathbb{R}^{r_k} \longrightarrow \mathcal{X}_{k-1} \times \mathcal{X}_k, \quad (18)$$

$$(t, \mathbf{z}_{k-1}^\ell) \longmapsto \begin{bmatrix} \hat{x}_{k-1}^{k,\ell}(t) \\ \hat{x}_k^{k,\ell}(t) \end{bmatrix},$$

whereas the parameter head satisfies

$$\mathcal{Q}_k : \mathbb{R}^{r_k} \longrightarrow \Theta_{k-1}, \quad (19)$$

$$\mathbf{z}_{k-1}^\ell \longmapsto \hat{\theta}_{k-1}^{k,\ell}.$$

Consequently, the complete trajectory-conditioned PINN is

$$\Phi^k : [0, T] \times \mathcal{U}_{k-1} \longrightarrow (\mathcal{X}_{k-1} \times \mathcal{X}_k) \times \Theta_{k-1}, \quad (20)$$

$$(t, \mathbf{U}_{k-1}^\ell) \longmapsto (\mathcal{S}_k(t, \mathbf{z}_{k-1}^\ell), \mathcal{Q}_k(\mathbf{z}_{k-1}^\ell)),$$

with parameter block $\Theta_{PINN}^k := (\Theta_{E_k}, \Theta_{S_k}, \Theta_{Q_k})$. Since the parameter head does not receive t , $\hat{\theta}_{k-1}^{k,\ell}$ is constant over $[0, T]$ by construction. The state decoder uses twice continuously differentiable activation functions, so that

$$\dot{\hat{x}}_{k-1}^{k,\ell}(t) = \frac{\partial}{\partial t} \mathcal{S}_{k,1}(t, \mathbf{z}_{k-1}^\ell) \quad (21)$$

is computed by automatic differentiation while $\mathbf{z}_{k-1}^\ell$ is held fixed.

A hatted quantity carries two indices: the subscript indicates the state or parameter coordinate, and the superscript indicates the cell that produced it. By convention, $\hat{x}_1^2 \equiv \hat{x}_1$ denotes the reconstruction of the measured output. We write $\hat{x}_k := \hat{x}_k^k$ and $\hat{\theta}_{k-1} := \hat{\theta}_{k-1}^k$ when no ambiguity arises. Thus, the reported state trajectory is $\hat{\mathbf{x}} := (\hat{x}_1^2, \hat{x}_2^2, \dots, \hat{x}_n^2)$. The final cell also reconstructs $\hat{x}_n^{n+1}$, but this is an auxiliary trajectory used in the last physics residual and in the disturbance-coefficient estimate; it is not a second reported estimate of x_n . For readability, the trajectory index ℓ is suppressed in the loss expressions below; the reported training loss is the empirical average of these trajectory-wise losses over $\ell = 1, \dots, P$.

Finite-horizon interpretation. Because $\mathbf{U}_{k-1}^\ell$ contains measurements sampled over the complete interval $[0, T]$, the resulting PIBE is a finite-horizon, trajectory-level estimator and is not causal in time. A causal ver-

sion would replace $\mathbf{U}_{k-1}^\ell$ by a past-only sliding window; that extension is outside the scope of this work.

The local objective of the k -th PINN is denoted $\mathcal{L}_{Loc}^k$ and combines a data-fit term with a physics term. The weighted objective $\mathcal{L}_{Tot}^k$ is used only during joint fine-tuning, as detailed below. The steps of the PIBE are:

-Step 1: for $k = 2$, the sampled measurement trajectory $\mathbf{U}_1 = \mathbf{Y}$ is passed through $\mathcal{E}_2$. The decoder $\mathcal{S}_2$ reconstructs $(\hat{x}_1^2(t), \hat{x}_2^2(t))$, while the time-independent head $\mathcal{Q}_2$ estimates the constant parameter $\hat{\theta}_1^2$. The parameter block Θ_{PINN}^2 is optimized using the objective

$$\mathcal{L}_{Loc}^2(\Theta_{PINN}^2) = \mathcal{L}_{Data}^2(y, \hat{x}_1^2 | \Theta_{PINN}^2) + \lambda \mathcal{L}_{Physics}^2(\hat{x}_1^2, \hat{x}_2^2, \hat{\theta}_1^2 | \Theta_{PINN}^2), \quad (22)$$

where the data term measures the mismatch between the measurement and the PINN reconstruction,

$$\mathcal{L}_{Data}^2(y, \hat{x}_1^2 | \Theta_{PINN}^2) = \frac{1}{N} \sum_{i=1}^N (y(t_i) - \hat{x}_1^2(t_i))^2, \quad (23)$$

and the physics term is the squared residual of the first ODE $\dot{x}_1 = x_2 + f_1(x_1, \theta_1)$ evaluated on the PINN outputs:

$$\mathcal{L}_{Physics}^2(\hat{x}_1^2, \hat{x}_2^2, \hat{\theta}_1^2 | \Theta_{PINN}^2) = \frac{1}{N_r} \sum_{j=1}^{N_r} |r_2(\tau_j)|^2, \quad (24)$$

with the residual

$$r_2(t) := \dot{\hat{x}}_1^2(t) - \hat{x}_2^2(t) - f_1(\hat{x}_1^2(t), \hat{\theta}_1^2). \quad (25)$$

By construction, r_2 is a function of the $k = 2$ PINN outputs and the known nonlinearity f_1 .

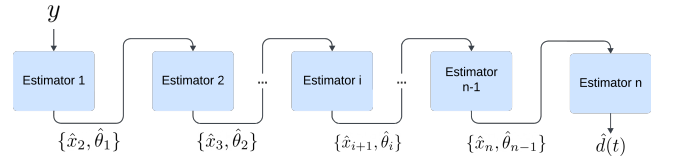


Fig. 2. Block diagram of the PINN-based chain of estimators

-Step 2: for $k = 3, \dots, n$, the encoder receives the

trajectory-level input $\mathbf{U}_{k-1}$ defined in (15). The corresponding continuous upstream arguments are grouped as

$$\begin{aligned}\mathcal{I}_{k-2}^x(t) &:= \text{col}\left(\hat{x}_1^2(t), \{\hat{x}_j^j(t)\}_{j=2}^{k-2}\right), \\ \mathcal{I}_{k-2}^\theta &:= \text{col}\left(\{\hat{\theta}_j^{j+1}\}_{j=1}^{k-2}\right), \\ \mathcal{I}_{k-1}(t) &:= (\mathcal{I}_{k-2}^x(t), \mathcal{I}_{k-2}^\theta).\end{aligned}\quad (26)$$

Empty indexed sequences are omitted. Thus, $\mathcal{I}_{k-2}^x$ contains precisely the upstream state arguments $x_1, \dots, x_{k-2}$, while $\mathcal{I}_{k-2}^\theta$ contains precisely $\theta_1, \dots, \theta_{k-2}$. These quantities are held fixed during local pre-training and are used to evaluate the physics residual. The k -th cell estimates $x_k(t)$ and the constant parameter θ_{k-1} . To incorporate the previous estimations into the loss of the current estimator, we propose the weighted global loss

$$\mathcal{L}_{Tot}^k(\mathfrak{W}_{2:k}) = \sum_{i=2}^k e^{-\frac{k-i}{k}} \mathcal{L}_{Loc}^i(\Theta_{PINN}^i), \quad (27)$$

where $\mathfrak{W}_{2:k} := \{\Theta_{PINN}^i\}_{i=2}^k$ collects all PINN parameters up to step k , and the local loss $\mathcal{L}_{Loc}^k$ is associated with the model estimating (x_k, θ_{k-1}) and defined by

$$\begin{aligned}\mathcal{L}_{Loc}^k(\Theta_{PINN}^k) &= \mathcal{L}_{Data}^k(\hat{x}_{k-1}^{k-1}, \hat{x}_{k-1}^k | \Theta_{PINN}^k) \\ &+ \lambda \mathcal{L}_{Physics}^k(\mathcal{I}_{k-1}, \hat{x}_{k-1}^k, \hat{x}_k^k, \hat{\theta}_{k-1}^k | \Theta_{PINN}^k),\end{aligned}\quad (28)$$

with the consistency-with-upstream data term

$$\mathcal{L}_{Data}^k(\hat{x}_{k-1}^{k-1}, \hat{x}_{k-1}^k | \Theta_{PINN}^k) = \frac{1}{N} \sum_{i=1}^N (\hat{x}_{k-1}^{k-1}(t_i) - \hat{x}_{k-1}^k(t_i))^2, \quad (29)$$

and the physics term given by the squared residual of the $(k-1)$ -th ODE,

$$\mathcal{L}_{Physics}^k(\dots | \Theta_{PINN}^k) = \frac{1}{N_r} \sum_{j=1}^{N_r} |r_k(\tau_j)|^2, \quad (30)$$

where the residual is

$$\begin{aligned}r_k(t) &:= \hat{x}_{k-1}^k(t) - \hat{x}_k^k(t) \\ &- f_{k-1}(\mathcal{I}_{k-2}^x(t), \hat{x}_{k-1}^k(t), \mathcal{I}_{k-2}^\theta, \hat{\theta}_{k-1}^k).\end{aligned}\quad (31)$$

The arguments of f_{k-1} in (31) are therefore ordered as

$$(\hat{x}_1^2, \dots, \hat{x}_{k-2}^{k-2}, \hat{x}_{k-1}^k, \hat{\theta}_1^2, \dots, \hat{\theta}_{k-2}^{k-1}, \hat{\theta}_{k-1}^k),$$

exactly matching its signature. Thus, r_k depends only on (i) the frozen upstream estimates in $\mathcal{I}_{k-1}$ and (ii) the current PINN outputs $\hat{x}_{k-1}^k$, $\hat{x}_k^k$, and $\hat{\theta}_{k-1}^k$; no unknown quantities appear. Comparing the two estimators' outputs through $\mathcal{L}_{Data}^k$ enforces consistency at the x_{k-1} coordinate and enables fine-tuning of upstream weights once gradients are propagated through the global loss (27).

-Step 3: After estimating (x_k, θ_{k-1}) for $k = 2, \dots, n$, we introduce a final cell, indexed by $n+1$, to estimate the structured disturbance coefficients. For trajectory ℓ , its

encoder first computes

$$\mathbf{z}_n^\ell = \mathcal{E}_{n+1}(\mathbf{U}_n^\ell; \Theta_{E_{n+1}}) \in \mathbb{R}^{r_{n+1}}, \quad (32)$$

where $\mathbf{U}_n^\ell$ contains the measurement trajectory, all upstream state estimates, and all parameter estimates, according to (15). The final state decoder and coefficient head are defined by

$$\begin{aligned}\mathcal{S}_{n+1} : [0, T] \times \mathbb{R}^{r_{n+1}} &\longrightarrow \mathcal{X}_n, \\ (t, \mathbf{z}_n^\ell) &\longmapsto \hat{x}_n^{n+1, \ell}(t),\end{aligned}\quad (33)$$

and

$$\begin{aligned}\mathcal{Q}_{n+1} : \mathbb{R}^{r_{n+1}} &\longrightarrow \mathcal{A}, \\ \mathbf{z}_n^\ell &\longmapsto \hat{a}^\ell.\end{aligned}\quad (34)$$

The output of $\mathcal{Q}_{n+1}$ is constrained to the admissible set $\mathcal{A}$, for instance through an output reparameterization when $\mathcal{A}$ is a box. The complete final cell is therefore

$$\begin{aligned}\Phi^{n+1} : [0, T] \times \mathcal{U}_n &\longrightarrow \mathcal{X}_n \times \mathcal{A}, \\ (t, \mathbf{U}_n^\ell) &\longmapsto (\mathcal{S}_{n+1}(t, \mathbf{z}_n^\ell), \mathcal{Q}_{n+1}(\mathbf{z}_n^\ell)),\end{aligned}\quad (35)$$

with

$$\Theta_{PINN}^{n+1} := (\Theta_{E_{n+1}}, \Theta_{S_{n+1}}, \Theta_{Q_{n+1}}).$$

Moreover,

$$\dot{\hat{x}}_n^{n+1, \ell}(t) = \frac{\partial}{\partial t} \mathcal{S}_{n+1}(t, \mathbf{z}_n^\ell) \quad (36)$$

is computed by automatic differentiation while $\mathbf{z}_n^\ell$ is held fixed. Thus, $\hat{a}^\ell$ may differ between trajectories but is constant with respect to t within each trajectory. As above, we suppress ℓ in the loss expressions below. Define

$$\hat{\mathbf{x}}^{n+1}(t) := \text{col}\left(\hat{x}_1^2(t), \{\hat{x}_j^j(t)\}_{j=2}^{n-1}, \hat{x}_n^{n+1}(t)\right).$$

The middle sequence is empty when $n = 2$. This is the auxiliary state vector used only to evaluate the final-cell physics residual. Define also

$$\hat{\boldsymbol{\theta}} := (\hat{\theta}_1^2, \dots, \hat{\theta}_{n-1}^n).$$

The local loss of the final cell is

$$\mathcal{L}_{Loc}^{n+1}(\Theta_{PINN}^{n+1}) = \mathcal{L}_{Data}^{n+1} + \lambda \mathcal{L}_{Physics}^{n+1}, \quad (37)$$

where the consistency term is

$$\mathcal{L}_{Data}^{n+1} = \frac{1}{N} \sum_{i=1}^N |\hat{x}_n^n(t_i) - \hat{x}_n^{n+1}(t_i)|^2, \quad (38)$$

and the physics term associated with the last ODE is

$$\mathcal{L}_{Physics}^{n+1} = \frac{1}{N_r} \sum_{j=1}^{N_r} |r_{n+1}(\tau_j)|^2, \quad (39)$$

with

$$\begin{aligned}r_{n+1}(t) &:= \dot{\hat{x}}_n^{n+1}(t) - f_n(\hat{\mathbf{x}}^{n+1}(t), \hat{\boldsymbol{\theta}}) \\ &- \Gamma_q(t)^\top \hat{a}.\end{aligned}\quad (40)$$

The resulting disturbance estimate is

$$\hat{d}(t) := \Gamma_q(t)^\top \hat{a}. \quad (41)$$

Thus, $\hat{d}$ estimates the structured component of (2), while r_q represents the irreducible finite-basis approximation error. After local pre-training, all cells are jointly fine-

tuned using the extended global loss

$$\mathcal{L}_{Tot}^{n+1} = \sum_{i=2}^{n+1} e^{-\frac{n+1-i}{n+1}} \mathcal{L}_{Loc}^i. \quad (42)$$

The general estimation process is summarized in Algorithm 1.

Algorithm 1 Training Procedure for the PIBE framework.

Input: Sampled trajectories $\{y^\ell(t_i) : \ell = 1, \dots, P, i = 1, \dots, N\}$; dynamical system model (1); per-cell iteration budget N_{tot} ; local pre-training cutoff N_{par} satisfying $0 < N_{par} < N_{tot}$; weighting factor λ

Output: Trained trajectory-conditioned cells $\{\Phi^k\}_{k=2}^{n+1}$; estimates $\{\hat{x}, \hat{\theta}, \hat{a}, \hat{d}\}$, with $\hat{d}(t) = \Gamma_q(t)^\top \hat{a}$
All losses are averaged over the selected trajectory mini-batch.

```

1: for  $k \leftarrow 2$  to  $n+1$  do
2:   Initialize the current cell parameters  $\Theta_{PINN}^k$ 
3:   for  $i \leftarrow 0$  to  $N_{tot} - 1$  do
4:     Select a mini-batch of trajectory indices
5:     if  $i < N_{par}$  then ▷ local pre-training
6:       Freeze  $\{\Theta_{PINN}^m\}_{m=2}^{k-1}$ 
7:       Construct  $\mathbf{U}_{k-1}$  from detached upstream
       outputs, with  $\mathbf{U}_1 = \mathbf{Y}$  when  $k = 2$ 
8:       Evaluate  $\Phi^k$  and compute  $\mathcal{L}_{Loc}^k$  using
       (22), (28), or (37), according to  $k$ 
9:       Update only  $\Theta_{PINN}^k$  using  $\nabla_{\Theta_{PINN}^k} \mathcal{L}_{Loc}^k$ 
10:    else ▷ end-to-end fine-tuning
11:      Unfreeze  $\{\Theta_{PINN}^m\}_{m=2}^k$ 
12:      Evaluate  $\{\Phi^m\}_{m=2}^k$  sequentially without
       detaching the upstream outputs
13:      Compute  $\{\mathcal{L}_{Loc}^m\}_{m=2}^k$ 
14:       $\mathcal{L}_{Tot}^k \leftarrow \sum_{m=2}^k e^{-(k-m)/k} \mathcal{L}_{Loc}^m$ 
15:      Update  $\{\Theta_{PINN}^m\}_{m=2}^k$  using
        $\nabla_{\{\Theta_{PINN}^m\}_{m=2}^k} \mathcal{L}_{Tot}^k$ 
16:    end if
17:  end for
18: end for
19: Set  $\hat{d}(t) \leftarrow \Gamma_q(t)^\top \hat{a}$ 

```

3.2 Proposed Energy-Based Physics Informed bank of estimators: Non-Gaussian Noise

We now focus on the compactly supported non-Gaussian case. Mean-squared error targets a conditional mean irrespective of whether the noise is Gaussian; non-Gaussianity alone therefore does not create bias. However, an unknown nonzero noise mean shifts that conditional mean, and a quadratic data term alone cannot separate the physical signal from an unknown location shift. To model residual shapes without prescribing a named parametric family, we introduce the Energy-Based PINN Bank of Estimators (EPIBE), which combines PINNs with scalar EBMs following [27]. The

identification conditions needed for this combination are made explicit below.

PINN and Non-Gaussian Noise: Consider a general nonlinear ODE

$$F(t, y, \dot{y}, \ddot{y}, \dots, y^{(n)}) = 0. \quad (43)$$

and noisy observations

$$y^{\text{obs}}(t) = y(t) + \omega(t), \quad \mathbb{E}[\omega(t)] = \mu_\omega, \quad \text{Var}(\omega(t)) = \sigma_\omega^2.$$

For a candidate trajectory $\hat{y}$ and a candidate scalar location m , define the population quadratic data risk

$$\mathcal{R}_D(\hat{y}, m) := \frac{1}{T} \int_0^T \mathbb{E}[(\hat{y}(t) + m - y^{\text{obs}}(t))^2] dt. \quad (44)$$

Proposition 1 (Location ambiguity of the quadratic data term) Assume the noise has the time-independent first two moments specified above. Then

$$\mathcal{R}_D(\hat{y}, m) = \sigma_\omega^2 + \frac{1}{T} \int_0^T (\hat{y}(t) - y(t) + m - \mu_\omega)^2 dt. \quad (45)$$

Consequently, the data risk alone is minimized by every pair satisfying

$$\hat{y}(t) - y(t) = \mu_\omega - m \quad \text{for a.e. } t \in [0, T], \quad (46)$$

and therefore does not separately identify y and μ_ω . If a nonnegative population physics functional $\mathcal{R}_P(\hat{y})$ has the unique zero $\hat{y} = y$ over the admissible trajectory class, then, for every $\lambda > 0$, the combined risk

$$\mathcal{R}_D(\hat{y}, m) + \lambda \mathcal{R}_P(\hat{y})$$

has the unique minimizer $(\hat{y}, m) = (y, \mu_\omega)$.

Proof 1 Write $\omega = \mu_\omega + (\omega - \mu_\omega)$, expand the square in (44), and use $\mathbb{E}[\omega - \mu_\omega] = 0$. This gives (45). Its smallest possible value is σ_ω^2 , attained precisely by (46). The true pair (y, μ_ω) attains this value and has zero physics risk. Any other pair in the data-risk equivalence class has strictly positive physics risk under the unique-zero hypothesis, while a pair outside that class has strictly larger data risk. $\square$

Remark 2 (What the proposition does and does not establish) The unique-zero condition is a coercivity or identifiability requirement; it is not supplied merely by adding a finite physics-loss weight. In the proposed state-parameter setting, its role is played conditionally by Assumption 2 together with the defect-to-solution stability qualification in Remark 10. Moreover, EPIBE estimates a complete residual density rather than an independent scalar m . A small EBM negative log-likelihood alone does not prove that the PINN output is the physical signal: the EBM can otherwise absorb a systematic PINN error as part of the residual law.

Assumption 3 (Well-posed scalar EBM class)

For every cell $k = 2, \dots, n+1$, fix a compact residual

support $\mathcal{R}_k = [-\bar{\rho}_k, \bar{\rho}_k]$ containing all admissible training and inference residuals. The radius is fixed before EBM activation using the compact decoder-output and parameter ranges defined above together with the known noise-support bound; the warm-up residual range may be used only as a diagnostic. Define the gauge-normalized energy

$$\tilde{E}_{\zeta_k}(\xi) := E_{\zeta_k}(\xi) - E_{\zeta_k}(0). \quad (47)$$

The EBM parameter set $\mathcal{K}_k$ is compact, the map $\zeta_k \mapsto \tilde{E}_{\zeta_k}$ is continuous from $\mathcal{K}_k$ into $C(\mathcal{R}_k)$ equipped with the uniform norm, and there exist finite constants $B_{E,k}$ and $L_{E,k}$ such that, for every $\zeta_k \in \mathcal{K}_k$,

$$\|\tilde{E}_{\zeta_k}\|_{L^\infty(\mathcal{R}_k)} \leq B_{E,k}, \quad \text{Lip}_{\mathcal{R}_k}(\tilde{E}_{\zeta_k}) \leq L_{E,k}. \quad (48)$$

These constraints can be implemented by bounded output parameterizations, spectral normalization, and projected parameter updates.

Under Assumption 3, the k -th residual density is

$$p_{k,\zeta_k}(\xi) := \frac{\exp[-\beta \tilde{E}_{\zeta_k}(\xi)]}{Z_k(\tilde{E}_{\zeta_k})} \mathbf{1}_{\mathcal{R}_k}(\xi), \quad (49)$$

$$Z_k(\tilde{E}_{\zeta_k}) := \int_{\mathcal{R}_k} \exp[-\beta \tilde{E}_{\zeta_k}(u)] du.$$

The energy bound gives the explicit estimates

$$2\bar{\rho}_k e^{-\beta B_{E,k}} \leq Z_k(\tilde{E}_{\zeta_k}) \leq 2\bar{\rho}_k e^{\beta B_{E,k}},$$

$$\frac{e^{-2\beta B_{E,k}}}{2\bar{\rho}_k} \leq p_{k,\zeta_k}(\xi) \leq \frac{e^{2\beta B_{E,k}}}{2\bar{\rho}_k}, \quad \xi \in \mathcal{R}_k. \quad (50)$$

The gauge normalization does not change the represented density, while (50) prevents arbitrarily narrow empirical-likelihood spikes. For fixed admissible residual samples, the EBM subproblem therefore has a finite, bounded-below objective and, by compactness and continuity, attains a minimum. This statement does not establish existence of a minimizer for the unrestricted joint PINN–EBM problem, statistical consistency, or the physical identifiability discussed in Remark 2.

Proposition 2 (*Conditional population interpretation of the EBM*) Fix a cell k and a specified pooling distribution over trajectories and sampling times. Suppose that the resulting scalar consistency residual has a pooled marginal density p_k^* supported on $\mathcal{R}_k$, with finite differential entropy. For a fixed, correctly identified physical signal, define the population EBM risk

$$\mathcal{R}_{E,k}(\zeta_k) := \mathbb{E}_{\xi \sim p_k^*}[-\log p_{k,\zeta_k}(\xi)].$$

Then

$$\mathcal{R}_{E,k}(\zeta_k) = h(p_k^*) + D_{\text{KL}}(p_k^* \| p_{k,\zeta_k}), \quad (51)$$

where $h(p_k^*)$ is independent of ζ_k . Consequently, if p_k^* belongs to the density class in (49), every population-risk minimizer represents p_k^* almost everywhere, although the minimizing network parameter need not be unique. If p_k^* is not in the class, any population-risk minimizer is instead a KL projection onto the admissible density class.

Proof 2 Add and subtract $-\mathbb{E}_{p_k^*}[\log p_k^*(\xi)]$ in the definition of $\mathcal{R}_{E,k}$. This yields (51). The conclusions follow from the nonnegativity of the Kullback–Leibler divergence and its equality condition. $\square$

Remark 3 (Scope of the learning claim) Proposition 2 is a population statement conditional on the physical signal. Passing from the empirical loss (53) to this population risk additionally requires an appropriate independent-sampling or mixing assumption, a uniform law of large numbers for the constrained energy class, and successful optimization. These properties are not established here. In particular, time samples along a single deterministic trajectory cannot be treated as independent residual draws without an explicit stochastic sampling model. The proposition therefore justifies the NLL’s pooled-marginal density-fitting interpretation, but does not constitute a joint PINN–EBM consistency theorem. In particular, the unconditioned scalar EBM used here does not represent a time-, state-, or trajectory-conditioned residual law; such a heteroscedastic model would require conditioning the energy on the corresponding context.

We present now the proposed algorithm for the joint estimation of states, parameters, and disturbance for the system described in Eq. 1, with measurements affected by non-Gaussian noise. As outlined in Section 2, given the measured output $y(t)$, our objective is to estimate the states $x_k(t)$, $k = 2, \dots, n$ and the parameters θ_i , $i = 1, \dots, n-1$ and the disturbance $d(t)$. The estimation process follows similar steps as in section 3. Some technical modifications are given below.

-Step 1: for $k = 2$, we use the noisy output measurements $y(t)$ to estimate the state $x_2(t)$ and the parameter θ_1 through a combined PINN and EBM architecture. The PINN component is the trajectory-conditioned cell Φ^2 defined previously. The EBM models the scalar residual $\varepsilon = y - \hat{x}_1^2$. This residual equals the measurement noise only when the physical signal has been correctly reconstructed; otherwise, it also contains the first-cell reconstruction error. After the PINN-only warm-up described below, the PINN and EBM parameters are updated jointly using the local objective

$$\mathcal{L}_{\text{Loc},E}^2 = \mathcal{L}_{\text{Data}}^{\text{ebm},2}(\{y - \hat{x}_1^2\} | \Theta_{\text{PINN}}^2, \Theta_{\text{EBM}}^2) + \lambda \mathcal{L}_{\text{Physics}}^2(\hat{x}_1^2, \hat{x}_2^2, \hat{\theta}_1^2 | \Theta_{\text{PINN}}^2), \quad (52)$$

where

$$\mathcal{L}_{\text{Data}}^{\text{ebm},k}(\{\varepsilon_k\} | \Theta_{\text{PINN}}^k, \Theta_{\text{EBM}}^k) = \log Z_k(\tilde{E}_{\Theta_{\text{EBM}}^k}) + \frac{\beta}{N} \sum_{i=1}^N \tilde{E}_{\Theta_{\text{EBM}}^k}(\varepsilon_k(t_i)), \quad (53)$$

for $k = 2, \dots, n+1$, with the corresponding cell consistency residual ε_k . In particular, $\varepsilon_2 = y - \hat{x}_1^2$. Every residual entering (53) is constrained to $\mathcal{R}_k$; otherwise its

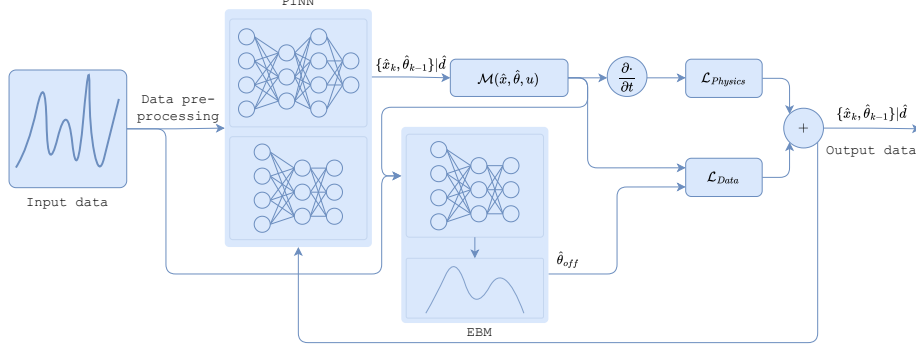


Fig. 3. A unitary estimation cell of the proposed Energy-based Physics-Informed Bank of Estimators (EPIBE) estimator.

likelihood under (49) is zero. The mean associated with the learned first-cell residual density is the descriptive moment

$$\hat{\mu}_\omega := \int_{\mathcal{R}_2} \xi p_{2, \Theta_{EBM}^2}(\xi) d\xi. \quad (54)$$

It is not an independently optimized offset and equals the true noise mean only when the physical signal is correctly identified and the learned first-cell residual density is statistically consistent.

-Step 2: for $k = 3, \dots, n$, we follow the same recursive procedure as in Section 3.1, replacing the quadratic consistency term by its energy-based counterpart. The corresponding local loss is

$$\begin{aligned} \mathcal{L}_{Loc,E}^k &= \mathcal{L}_{Data}^{ebm,k}(\{\hat{x}_{k-1}^{k-1} - \hat{x}_{k-1}^k\} | \Theta_{PINN}^k, \Theta_{EBM}^k) \\ &+ \lambda \mathcal{L}_{Physics}^k(\mathcal{I}_{k-1}, \hat{x}_{k-1}^k, \hat{x}_k^k, \hat{\theta}_{k-1}^k | \Theta_{PINN}^k), \end{aligned} \quad (55)$$

where $\hat{x}_{k-1}^{k-1}(t)$ is the estimate supplied by the previous cell and $\hat{x}_{k-1}^k(t)$ is the reconstruction produced by the current cell. Their difference is the scalar consistency residual modeled by the k -th EBM. Because the measurement enters only the first coordinate, how its uncertainty propagates to the downstream states is unknown; we therefore fit a separate residual model at each cell rather than assuming that the original measurement-noise law is preserved throughout the bank. During end-to-end fine-tuning, the energy-based weighted loss is

$$\mathcal{L}_{Tot,E}^k = \sum_{m=2}^k e^{-\frac{k-m}{k}} \mathcal{L}_{Loc,E}^m, \quad k = 2, \dots, n. \quad (56)$$

-Step 3: The EPIBE uses the same final cell (35), coefficient vector $\hat{a}$, physics residual (40), and disturbance estimate (41) as the PIBE. Only its consistency term is replaced by the energy-based data loss, giving

$$\begin{aligned} \mathcal{L}_{Loc,E}^{n+1} &= \mathcal{L}_{Data}^{ebm,n+1}(\{\hat{x}_n^n - \hat{x}_n^{n+1}\} | \Theta_{PINN}^{n+1}, \Theta_{EBM}^{n+1}) \\ &+ \lambda \mathcal{L}_{Physics}^{n+1}, \end{aligned} \quad (57)$$

where $\mathcal{L}_{Physics}^{n+1}$ is defined in (39). The parameters

Θ_{PINN}^{n+1} , including the coefficient-head parameters $\Theta_{Q_{n+1}}$, and Θ_{EBM}^{n+1} are trained jointly after the warm-up phase, followed by global fine-tuning of all upstream cells using

$$\mathcal{L}_{Tot,E}^{n+1} = \sum_{m=2}^n e^{-\frac{n+1-m}{n+1}} \mathcal{L}_{Loc,E}^m + \mathcal{L}_{Loc,E}^{n+1}. \quad (58)$$

Remark 4 (Tractability of the partition function)

For general high-dimensional response variables, evaluating $Z(E_\zeta) = \int e^{-\beta E_\zeta(\tilde{y})} d\tilde{y}$ is intractable, and EBM training typically relies on Monte-Carlo schemes such as Langevin dynamics or contrastive divergence [7,25]. Here, every cell $k \in \{2, \dots, n+1\}$ models a scalar consistency residual on the fixed support $\mathcal{R}_k$ from Assumption 3. The one-dimensional partition function is evaluated as

$$Z_k(\tilde{E}_{\zeta_k}) = \int_{-\bar{\rho}_k}^{\bar{\rho}_k} e^{-\beta \tilde{E}_{\zeta_k}(\xi)} d\xi. \quad (59)$$

The radius must cover the a priori admissible residual range; a warm-up range with a safety margin alone is not a mathematical support guarantee. For $k = 2$, the bound accounts for $\tilde{w}$ and the compact decoder-output range; downstream radii also account for admissible upstream discrepancies. Equation (59) is computed at constant cost per gradient step by a fixed Gauss–Legendre rule on 10^2 – 10^3 nodes, and the same quadrature evaluates $\nabla_{\zeta_k} \log Z_k = -\beta \mathbb{E}_{\xi \sim p_{k,\zeta_k}}[\nabla_{\zeta_k} \tilde{E}_{\zeta_k}(\xi)]$ required to backpropagate through the NLL. When numerical quadrature is used, the implemented objective and gradient are approximations of (53); any statistical guarantee would therefore also have to control the quadrature error.

Remark 5 (Training schedule)

The joint PINN–EBM optimization is sensitive to initialization because the EBM is fitted to residuals produced by an initially untrained PINN. We therefore impose $0 < N_{EBM} < N_{par} < N_{tot}$ and use three phases for each cell. For $i < N_{EBM}$, the EBM parameters are frozen and only the current PINN is trained with the corresponding PIBE local objective, using a warm-up physics weight equal to one. For $N_{EBM} \leq i < N_{par}$, the current PINN

and its EBM are trained jointly with $\mathcal{L}_{Loc,E}^k$, while all upstream cells remain frozen. Finally, for $i \geq N_{par}$, all PINN and EBM parameters up to the current cell are fine-tuned end-to-end using the weighted energy-based global loss. Thus, $\hat{\mu}_\omega$ in (54) is a moment of the learned density, not a separately optimized parameter or a guaranteed estimate of μ_ω . Every EBM update is constrained to the class in Assumption 3; without this restriction, joint empirical likelihood optimization can collapse toward increasingly concentrated residual densities. In practice, N_{EBM} is selected so that the warm-up data and physics losses have stabilized before activating the EBM.

The general estimation process is summarized in the Algorithm 2.

4 Joint state, parameter, and disturbance error propagation analysis

In this section, we analyze error propagation through the proposed bank of estimators for the states x_k , parameters θ_{k-1} , and disturbance $d(t)$ jointly. Since each PINN produces approximations that are fed to subsequent cells, quantifying the cumulative error is essential. The final disturbance cell contains contributions from all upstream cells, although its numerical bound need not be the largest because the cell Lipschitz constants and reference scales may differ; see Remark 13.

4.1 Notation, definitions, and prerequisites

We analyze one trajectory ℓ at a time and suppress the index ℓ whenever no ambiguity arises. Let

$$\|\mathbf{v}\|_{N,2} := \left(\frac{1}{N} \sum_{i=1}^N |v_i|^2 \right)^{1/2}, \quad \mathbf{v} \in \mathbb{R}^N, \quad (60)$$

denote the normalized discrete L^2 norm. To compare sampled states and scalar parameters without introducing a spurious dependence on the number N of time samples, fix positive reference scales s_y , s_{x_j} , s_{θ_j} , and s_a . We set $s_{x_1} := s_y$. For the trajectory vector $\mathbf{U}_{k-1}$ in (15), define

$$\begin{aligned} \|\mathbf{U}_{k-1}\|_{\mathcal{U}_{k-1}}^2 &:= \frac{\|\mathbf{Y}\|_{N,2}^2}{s_y^2} + \sum_{j=2}^{k-1} \frac{\|\mathbf{X}_j\|_{N,2}^2}{s_{x_j}^2} \\ &\quad + \sum_{j=1}^{k-2} \frac{|\theta_j|^2}{s_{\theta_j}^2}, \end{aligned} \quad (61)$$

where empty sums are zero. Equivalently, $\|\mathbf{U}_{k-1}\|_{\mathcal{U}_{k-1}} = \|\mathbf{D}_{k-1}\mathbf{U}_{k-1}\|_2$, where $\mathbf{D}_{k-1}$ divides every sampled-trajectory block by $\sqrt{N}$ and its reference scale, and every parameter block by its reference scale. This fixed normalization can be implemented as an input preprocessing layer, or absorbed into the first encoder layer. For nondimensionalized variables, all reference scales

Algorithm 2 Training Procedure for the EPIBE.

Input: Sampled trajectories $\{y^\ell(t_i) : \ell = 1, \dots, P, i = 1, \dots, N\}$; dynamical system model (1); per-cell iteration budget N_{tot} ; local pre-training cutoff N_{par} ; per-cell EBM activation iteration N_{EBM} satisfying $0 < N_{EBM} < N_{par} < N_{tot}$; weighting factor λ ; inverse temperature $\beta > 0$; fixed residual supports $\{\mathcal{R}_k\}_{k=2}^{n+1}$ and constrained energy classes from Assumption 3; fixed Gauss–Legendre quadrature rule

Output: Trained trajectory-conditioned cells $\{\Phi^k\}_{k=2}^{n+1}$ and energy models; estimates $\{\hat{x}, \hat{\theta}, \hat{a}, \hat{d}\}$, with $\hat{d}(t) = \Gamma_q(t)^\top \hat{a}$
All losses are averaged over the selected trajectory mini-batch.

```

1: for  $k \leftarrow 2$  to  $n+1$  do
2:   Initialize  $\Theta_{PINN}^k$  and  $\Theta_{EBM}^k \in \mathcal{K}_k$ 
3:   for  $i \leftarrow 0$  to  $N_{tot} - 1$  do
4:     Select a mini-batch of trajectory indices
5:     if  $i < N_{EBM}$  then  $\triangleright$  PINN-only warm-up
6:       Freeze all EBM parameters and all upstream cells
7:       Construct  $\mathbf{U}_{k-1}$  from detached upstream outputs, with  $\mathbf{U}_1 = \mathbf{Y}$  when  $k = 2$ 
8:       Evaluate  $\Phi^k$ 
9:       Compute the PIBE loss  $\mathcal{L}_{Loc}^k$  with warm-up physics weight 1
10:      Update only  $\Theta_{PINN}^k$ 
11:      else if  $i < N_{par}$  then  $\triangleright$  local PINN–EBM training
12:        Freeze all upstream PINN and EBM parameters
13:        Unfreeze  $\Theta_{PINN}^k$  and  $\Theta_{EBM}^k$ 
14:        Construct  $\mathbf{U}_{k-1}$  from detached upstream outputs, with  $\mathbf{U}_1 = \mathbf{Y}$  when  $k = 2$ 
15:        Evaluate  $\Phi^k$ 
16:        Compute  $\mathcal{L}_{Loc,E}^k$  using (52), (55), or (57), according to  $k$ 
17:        Jointly update  $\Theta_{PINN}^k$  and  $\Theta_{EBM}^k$ , projecting the EBM update onto  $\mathcal{K}_k$ 
18:      else  $\triangleright$  end-to-end PINN–EBM fine-tuning
19:        Unfreeze  $\{\Theta_{PINN}^m, \Theta_{EBM}^m\}_{m=2}^k$ 
20:        Evaluate  $\{\Phi^m\}_{m=2}^k$  sequentially without detaching the upstream outputs
21:        Compute  $\{\mathcal{L}_{Loc,E}^m\}_{m=2}^k$ 
22:         $\mathcal{L}_{Tot,E}^k \leftarrow \sum_{m=2}^k e^{-(k-m)/k} \mathcal{L}_{Loc,E}^m$ 
23:        Update  $\{\Theta_{PINN}^m, \Theta_{EBM}^m\}_{m=2}^k$  using the gradient of  $\mathcal{L}_{Tot,E}^k$ , with every EBM update constrained to its  $\mathcal{K}_m$ 
24:      end if
25:    end for
26:  end for
27: Set  $\hat{d}(t) \leftarrow \Gamma_q(t)^\top \hat{a}$ 

```

may be set to one. This removes the purely algebraic $\sqrt{N}$ factor; any claim uniform with respect to grid refinement additionally requires the encoder constants L_{E_k} to remain bounded as N changes.

For $j = 2, \dots, n$, let

$$\mathbf{X}_j^\ell := \left[x_j^\ell(t_1) \cdots x_j^\ell(t_N) \right]^\top \quad (62)$$

be the sampled true state. The exact-upstream input to cell k is

$$\mathbf{U}_{k-1}^{\star, \ell} := \text{col}(\mathbf{Y}^\ell, \mathbf{X}_2^\ell, \dots, \mathbf{X}_{k-1}^\ell, \theta_1, \dots, \theta_{k-2}), \quad (63)$$

whereas the input generated recursively by the trained upstream cells is

$$\check{\mathbf{U}}_{k-1}^\ell := \text{col}(\mathbf{Y}^\ell, \check{\mathbf{X}}_2^{2, \ell}, \dots, \check{\mathbf{X}}_{k-1}^{k-1, \ell}, \check{\theta}_1^{2, \ell}, \dots, \check{\theta}_{k-2}^{k-1, \ell}). \quad (64)$$

Here $\check{\mathbf{X}}_j^{j, \ell}$ contains the samples of $\check{x}_j^{j, \ell}$. Importantly, the same measured vector $\mathbf{Y}^\ell$ appears in (63) and (64). Thus, “exact-upstream” refers only to the state and parameter blocks; it does not mean that the measurement is noise free. In particular, $\mathbf{U}_1^{\star, \ell} = \check{\mathbf{U}}_1^\ell = \mathbf{Y}^\ell$.

To express the trained cell as an operator on complete trajectories, define, for $k = 2, \dots, n$,

$$\mathfrak{F}_k(\mathbf{U}) := \left(\mathcal{S}_k(\cdot, \mathcal{E}_k(\mathbf{U})), \mathcal{Q}_k(\mathcal{E}_k(\mathbf{U})) \right) \in \mathcal{Z}_k, \quad (65)$$

where $\mathcal{Z}_k := C([0, T]; \mathbb{R}^2) \times \mathbb{R}$. For the final cell,

$$\mathfrak{F}_{n+1}(\mathbf{U}) := \left(\mathcal{S}_{n+1}(\cdot, \mathcal{E}_{n+1}(\mathbf{U})), \mathcal{Q}_{n+1}(\mathcal{E}_{n+1}(\mathbf{U})) \right) \in \mathcal{Z}_{n+1}, \quad (66)$$

with $\mathcal{Z}_{n+1} := C([0, T]; \mathbb{R}) \times \mathbb{R}^q$. For $k = 2, \dots, n$, write $\mathbf{v} = (v_{k-1}, v_k)$ and define

$$\|\mathbf{v}\|_{\mathcal{S}_k} := \sup_{t \in [0, T]} \left(\frac{|v_{k-1}(t)|^2}{s_{x_{k-1}}^2} + \frac{|v_k(t)|^2}{s_{x_k}^2} \right)^{1/2}, \quad (67)$$

$$\|\vartheta\|_{\mathcal{Q}_k} := \frac{|\vartheta|}{s_{\theta_{k-1}}}.$$

For the final cell, set

$$\|v\|_{\mathcal{S}_{n+1}} := \frac{\|v\|_\infty}{s_{x_n}}, \quad \|b\|_{\mathcal{Q}_{n+1}} := \frac{\|b\|_2}{s_a}. \quad (68)$$

We equip the cell-output spaces with

$$\begin{aligned} \|(\mathbf{v}, \vartheta)\|_{\mathcal{Z}_k} &:= (\|\mathbf{v}\|_{\mathcal{S}_k}^2 + \|\vartheta\|_{\mathcal{Q}_k}^2)^{1/2}, \quad k = 2, \dots, n, \\ \|(v, b)\|_{\mathcal{Z}_{n+1}} &:= \left(\|v\|_{\mathcal{S}_{n+1}}^2 + \|b\|_{\mathcal{Q}_{n+1}}^2 \right)^{1/2}. \end{aligned} \quad (69)$$

These dimensionless norms make the input and output Lipschitz constants independent of the physical units used for individual states, parameters, and coefficients.

The exact-upstream and recursively generated cell out-

puts are

$$\bar{\mathbf{Z}}_k^\ell := \mathfrak{F}_k(\mathbf{U}_{k-1}^{\star, \ell}), \quad \check{\mathbf{Z}}_k^\ell := \mathfrak{F}_k(\check{\mathbf{U}}_{k-1}^\ell). \quad (70)$$

For $k = 2, \dots, n$, their new-state and parameter components are denoted $(\bar{x}_k^\ell, \bar{\theta}_{k-1}^\ell)$ and $(\check{x}_k^\ell, \check{\theta}_{k-1}^\ell)$, respectively. For $k = 2, \dots, n$, we denote the reconstructed-coordinate components by $\bar{x}_{k-1}^{k, \ell}$ and $\check{x}_{k-1}^{k, \ell}$, respectively. For $k = n+1$, the corresponding components are $(\bar{x}_n^{n+1, \ell}, \bar{a}^\ell)$ and $(\check{x}_n^{n+1, \ell}, \check{a}^\ell)$.

For $k = 2, \dots, n$, define the intrinsic and propagated errors by

$$\begin{aligned} \epsilon_{s_k}^{\text{int}}(t) &:= \bar{x}_k(t) - x_k(t), & \epsilon_{p_{k-1}}^{\text{int}} &:= \bar{\theta}_{k-1} - \theta_{k-1}, \\ \epsilon_{s_k}^{\text{prop}}(t) &:= \check{x}_k(t) - \bar{x}_k(t), & \epsilon_{p_{k-1}}^{\text{prop}} &:= \check{\theta}_{k-1} - \bar{\theta}_{k-1}. \end{aligned} \quad (71)$$

Consequently,

$$\begin{aligned} \check{x}_k - x_k &= \epsilon_{s_k}^{\text{prop}} + \epsilon_{s_k}^{\text{int}}, \\ \check{\theta}_{k-1} - \theta_{k-1} &= \epsilon_{p_{k-1}}^{\text{prop}} + \epsilon_{p_{k-1}}^{\text{int}}. \end{aligned} \quad (72)$$

When the superscript is omitted below, ϵ_{s_k} and $\epsilon_{p_{k-1}}$ refer to the corresponding intrinsic errors in (71). This decomposition isolates error created by the current trained cell from error inherited through its trajectory-level input. The former may include approximation, optimization, finite-data, and measurement effects; the latter is solely the effect of replacing $\mathbf{U}_{k-1}^\star$ by $\check{\mathbf{U}}_{k-1}$.

For the final cell, define $\epsilon_a^{\text{int}} := \bar{a} - a$ and $\epsilon_a^{\text{prop}} := \check{a} - \bar{a}$. Let

$$\begin{aligned} G_q &:= \sup_{t \in [0, T]} \|\Gamma_q(t)\|_2 < \infty, \\ \bar{d}_q(t) &:= \Gamma_q(t)^\top \bar{a}, \\ \check{d}_q(t) &:= \Gamma_q(t)^\top \check{a}. \end{aligned} \quad (73)$$

For notational compatibility, set $\bar{d} := \bar{d}_q$ and $\epsilon_d := \bar{d}_q - d$. Then

$$\begin{aligned} \|d - \bar{d}_q\|_\infty &\leq G_q \|\epsilon_a^{\text{int}}\|_2 + \varepsilon_{d, q}, \\ \|\check{d}_q - \bar{d}_q\|_\infty &\leq G_q \|\epsilon_a^{\text{prop}}\|_2. \end{aligned} \quad (74)$$

We use the following standard Lipschitz bound for multilayer networks.

Lemma 1 ([24]) *Let $\mathcal{N}$ be a neural network with L layers, weight matrices $A^{(j)}$, and Lipschitz-continuous activation functions $\varphi^{(j)}$ with constants α_j . Then $\mathcal{N}$ is Lipschitz continuous with constant $\Lambda = \prod_{j=1}^L \alpha_j \|A^{(j)}\|$.*

Assume that, on the admissible sets, the trained components satisfy

$$\begin{aligned} \|\mathcal{E}_k(\mathbf{U}) - \mathcal{E}_k(\tilde{\mathbf{U}})\|_2 &\leq L_{E_k} \|\mathbf{U} - \tilde{\mathbf{U}}\|_{\mathcal{Z}_{k-1}}, \\ \|\mathcal{S}_k(\cdot, z) - \mathcal{S}_k(\cdot, \tilde{z})\|_{\mathcal{S}_k} &\leq L_{S_k} \|z - \tilde{z}\|_2, \\ \|\mathcal{Q}_k(z) - \mathcal{Q}_k(\tilde{z})\|_{\mathcal{Q}_k} &\leq L_{Q_k} \|z - \tilde{z}\|_2. \end{aligned} \quad (75)$$

These constants can be bounded using Lemma 1; the scaling D_{k-1} is included in the encoder’s first layer when computing L_{E_k} , and the output-scale factors are included when computing L_{S_k} and L_{Q_k} . It follows directly

from (69) that the complete cell operator satisfies

$$\|\mathfrak{F}_k(\mathbf{U}) - \mathfrak{F}_k(\tilde{\mathbf{U}})\|_{\mathcal{Z}_k} \leq \Lambda_k \|\mathbf{U} - \tilde{\mathbf{U}}\|_{\mathcal{U}_{k-1}}, \quad (76)$$

where

$$\Lambda_k := L_{E_k} \sqrt{L_{S_k}^2 + L_{Q_k}^2}, \quad k = 2, \dots, n+1. \quad (77)$$

When the cell constants are uniformly bounded, for example through spectral normalization or Lipschitz-constrained training [26, 8], we write $\Lambda := \max_{2 \leq k \leq n+1} \Lambda_k$.

The bound (76) is conditional on the trained network parameters. It therefore applies to the PINN operators learned by either PIBE or EPIBE. For EPIBE, it does not by itself constitute a statistical consistency result for the learned energy model.

4.2 Assumptions

The deterministic propagation analysis uses the finite-horizon well-posedness and compact-admissibility condition in Assumption 1, together with the following assumptions specific to the cascaded PINN architecture. Assumption 2 is a model-level prerequisite for distinguishing the joint unknowns along exact model trajectories, but it is not used directly in the propagation recursion below. Connecting its local conditioning constant to the post-training intrinsic bounds requires the additional defect-to-solution stability estimate discussed in Remark 10.

For $k = 2, \dots, n$, define the ideal cell output

$$\mathbf{Z}_k^{\star, \ell} := \left(\begin{bmatrix} x_{k-1}^\ell(\cdot) \\ x_k^\ell(\cdot) \end{bmatrix}, \theta_{k-1} \right) \in \mathcal{Z}_k, \quad (78)$$

and, for the final cell, define

$$\mathbf{Z}_{n+1}^{\star, \ell} := (x_n^\ell(\cdot), a^\ell) \in \mathcal{Z}_{n+1}. \quad (79)$$

Assumption 4 (Intrinsic cell accuracy) *For every admissible trajectory ℓ , the trained cells evaluated at their exact-upstream inputs satisfy*

$$\|\bar{\mathbf{Z}}_k^\ell - \mathbf{Z}_k^{\star, \ell}\|_{\mathcal{Z}_k} \leq \bar{\varepsilon}_k^{\text{int}}, \quad k = 2, \dots, n+1, \quad (80)$$

for finite constants $\bar{\varepsilon}_k^{\text{int}}$. In particular, for $k = 2, \dots, n$ there exist component bounds

$$\begin{aligned} \|\bar{x}_k^\ell - x_k^\ell\|_\infty &\leq \bar{\varepsilon}_{s_k}, \\ |\bar{\theta}_{k-1}^\ell - \theta_{k-1}^\ell| &\leq \bar{\varepsilon}_{p_{k-1}}, \end{aligned} \quad \max \left\{ \frac{\bar{\varepsilon}_{s_k}}{s_{x_k}}, \frac{\bar{\varepsilon}_{p_{k-1}}}{s_{\theta_{k-1}}} \right\} \leq \bar{\varepsilon}_k^{\text{int}}, \quad (81)$$

and the final coefficient error satisfies

$$\|\bar{a}^\ell - a^\ell\|_2 \leq \bar{\varepsilon}_a \leq s_a \bar{\varepsilon}_{n+1}^{\text{int}}. \quad (82)$$

Consequently,

$$\|d^\ell - \bar{d}_q^\ell\|_\infty \leq G_q \bar{\varepsilon}_a + \varepsilon_{d,q}. \quad (83)$$

Assumption 4 is a post-training accuracy hypothesis, not a consequence of the Universal Approximation Theorem.

Universal approximation can support representability of continuous target maps on compact sets [10], but it does not guarantee identifiability, successful optimization, or a finite-sample error bound. Moreover, neither the state nor the disturbance error is asserted to vanish when measurement noise or the basis remainder r_q is nonzero. The result below determines precisely what can be inferred from one quadratic PIBE local loss. It yields a bound on a coupled state-parameter defect, but, as shown by an explicit counterexample, does not by itself verify the separate component bounds in (81). For EPIBE, a small negative log-likelihood does not by itself imply a small pointwise residual; an additional statistical calibration or concentration condition is required. The later deterministic propagation result can nevertheless be applied to either architecture conditionally on Assumption 4.

Assumption 5 (Temporal and grid regularity)

For every admissible input and every trained cell, each time-dependent component of $\mathcal{S}_k(t, \mathcal{E}_k(\mathbf{U}))$ is twice continuously differentiable in t . There exist constants $M_1, M_2 > 0$, independent of the trajectory and the cell index, such that

$$\begin{aligned} \sup_{\mathbf{U} \in \mathcal{U}_{k-1}} \sup_{t \in [0, T]} |\partial_t \mathcal{S}_{k,r}(t, \mathcal{E}_k(\mathbf{U}))| &\leq M_1, \\ \sup_{\mathbf{U} \in \mathcal{U}_{k-1}} \sup_{t \in [0, T]} |\partial_{tt} \mathcal{S}_{k,r}(t, \mathcal{E}_k(\mathbf{U}))| &\leq M_2. \end{aligned} \quad (84)$$

for every state-output coordinate r and every $k = 2, \dots, n+1$. The true trajectories satisfy $x_j \in W^{2,\infty}(0, T)$ for $j = 1, \dots, n-1$ and $x_n \in W^{1,\infty}(0, T)$, with uniform bounds over the admissible trajectory set. This regularity is compatible with $r_q \in L^\infty(0, T)$ and does not impose a nonexistent second derivative on x_n .

Let $r_k^\star$ denote the physics residual of cell k , evaluated at its exact-upstream output $\bar{\mathbf{Z}}_k$. There exist constants $L_{r_k} < \infty$, independent of the grids, such that

$$|r_k^\star(t) - r_k^\star(s)| \leq L_{r_k} |t - s|, \quad t, s \in [0, T], \quad k = 2, \dots, n+1. \quad (85)$$

For $k = 2, \dots, n$, the exact-upstream reconstruction error

$$e_{k-1}^{\text{rec}}(t) := \bar{x}_{k-1}^k(t) - x_{k-1}(t) \quad (86)$$

belongs to $W^{2,\infty}(0, T)$ and admits a uniform bound $\|\ddot{e}_{k-1}^{\text{rec}}\|_\infty \leq B_k$. For the final cell, no second derivative of x_n is assumed; the coefficient estimate will instead be treated in weak form by integrating the $\dot{x}_n$ error against the C^1 basis functions.

The ordered data and collocation grids include the endpoints of $[0, T]$. Their fill distances are

$$\begin{aligned} h_d &:= \sup_{t \in [0, T]} \min_{1 \leq i \leq N} |t - t_i|, \\ h_r &:= \sup_{t \in [0, T]} \min_{1 \leq j \leq N_r} |t - \tau_j|. \end{aligned} \quad (87)$$

and both grids are quasi-uniform: the ratio between the largest and smallest adjacent spacings is bounded inde-

pendently of N and N_r .

Assumption 5 contains no derivative-error convergence claim. Such a claim must be derived from the reconstruction loss, the grid coverage, and the uniform second-derivative bound; assuming it at this stage would make the later verification argument circular.

Parameter sensitivity. For each $k = 2, \dots, n$ and each time t , let $\xi_k(s, t)$, $s \in [0, 1]$, denote the segment joining the true pair $(x_{k-1}(t), \theta_{k-1})$ to the exact-upstream cell pair $(\bar{x}_{k-1}^k(t), \theta_{k-1})$, with all earlier state and parameter arguments held at their true values. We define the *mean-value sensitivities* of f_{k-1} with respect to x_{k-1} and θ_{k-1} by

$$a_k(t) := \int_0^1 \partial_{x_{k-1}} f_{k-1}(\xi_k(s, t)) ds, \quad (88)$$

$$\beta_k(t) := \int_0^1 \partial_{\theta_{k-1}} f_{k-1}(\xi_k(s, t)) ds. \quad (89)$$

Since $\xi_k(s, t)$ remains in the compact set on which f_{k-1} is C^1 , the pointwise bounds $|a_k(t)| \leq L_{f_{k-1}}^x$ and $|\beta_k(t)| \leq L_{f_{k-1}}^\theta$ hold.

Assumption 6 (Trajectory excitation) For each $k = 2, \dots, n$, the parameter θ_{k-1} is uniformly persistently excited over the horizon $[0, T]$: the time-averaged squared mean-value sensitivity is bounded below uniformly over all admissible error segments,

$$\gamma_k := \inf_{\xi_k(\cdot) \in \Xi_k} \frac{1}{T} \int_0^T \left(\int_0^1 \partial_{\theta_{k-1}} f_{k-1}(\xi_k(s, t)) ds \right)^2 dt > 0. \quad (90)$$

where Ξ_k is the set of segments $\xi_k(\cdot, \cdot)$ joining the true pair to any admissible estimated pair in the compact set (cf. (89)). Choosing the degenerate segment at the true pair recovers the usual single-trajectory excitation integral. The uniform infimum in (90) is strictly stronger than excitation only along the true trajectory; a local version may instead be imposed on a prescribed neighborhood when an a priori error radius is available.

Remark 6 Condition (90) is the continuous-time analog of a scalar regressor Gramian being uniformly positive [31, 11, 21]. The trajectory-specific integral can be evaluated a posteriori from an estimated trajectory, whereas certification of the uniform infimum requires a lower bound over the entire prescribed segment set Ξ_k . If this lower bound vanishes, the conditional estimate below provides no parameter guarantee; this loss of a first-order guarantee should not, by itself, be equated with a proof of global structural non-identifiability.

For $k = 2, \dots, n$, define the exact-upstream data and

physics losses by

$$\begin{aligned} \mathcal{L}_D^{k,*} &:= \frac{1}{N} \sum_{i=1}^N |g_{k,i}^*|^2, \\ \mathcal{L}_P^{k,*} &:= \frac{1}{N_r} \sum_{j=1}^{N_r} |r_k^*(\tau_j)|^2, \end{aligned} \quad (91)$$

where

$$g_{k,i}^* := \begin{cases} \bar{x}_1^2(t_i) - y(t_i), & k = 2, \\ \bar{x}_{k-1}^k(t_i) - x_{k-1}(t_i), & k = 3, \dots, n. \end{cases} \quad (92)$$

For $k = 2$, the exact-upstream loss equals the actual loss because $\mathbf{U}_1^* = \mathbf{Y}$. For $k \geq 3$, it is a theoretical counterfactual loss; relating it to the recursively evaluated training loss requires the propagation analysis below.

Lemma 2 (From a sampled mean to a uniform bound)

Let $g : [0, T] \rightarrow \mathbb{R}$ be L_g -Lipschitz, and let $\{s_i\}_{i=1}^M$ be a quasi-uniform grid with fill distance h . There exists $C_{\text{mesh}} > 0$, depending only on the mesh-ratio bound, such that

$$\frac{1}{T} \|g\|_{L^2(0,T)}^2 \leq C_{\text{mesh}} \left(\frac{1}{M} \sum_{i=1}^M |g(s_i)|^2 + L_g^2 h^2 \right). \quad (93)$$

Consequently, if the sampled mean is at most ε , then

$$\|g\|_\infty \leq \Psi_T(\varepsilon, h, L_g), \quad (94)$$

where

$$\Psi_T(\varepsilon, h, L) := \inf_{0 < \rho \leq T/2} \left\{ \sqrt{\frac{T}{\rho}} \sqrt{C_{\text{mesh}}(\varepsilon + L^2 h^2)} + L\rho \right\}. \quad (95)$$

In particular, $\Psi_T(\varepsilon, h, L) \rightarrow 0$ as $\varepsilon, h \rightarrow 0$ for fixed L .

Proof 3 Associate with each grid point its Voronoi interval. Quasi-uniformity implies that every such interval has length at most CT/M , with C depending only on the mesh ratio. On the interval associated with s_i , $|g(t)|^2 \leq 2|g(s_i)|^2 + 2L_g^2 h^2$. Integrating over all intervals gives (93), after enlarging the mesh constant if necessary. For any $t \in [0, T]$ and any $0 < \rho \leq T/2$, one can choose a one-sided interval $I_t \subset [0, T]$ of length ρ containing t . Lipschitz continuity and Cauchy-Schwarz give

$$|g(t)| \leq \rho^{-1/2} \|g\|_{L^2(I_t)} + L_g \rho.$$

Combining this inequality with (93) and minimizing over ρ proves (94). $\square$

For later use, define the derivative-recovery modulus

$$\mathcal{J}_T(D, B) := \inf_{0 < \rho \leq T/2} \left(\frac{2D}{\rho} + \frac{B\rho}{2} \right). \quad (96)$$

Taylor's formula on a suitable one-sided interval shows that every $e \in W^{2,\infty}(0, T)$ satisfying $\|e\|_\infty \leq D$ and $\|\ddot{e}\|_\infty \leq B$ obeys $\|\dot{e}\|_\infty \leq \mathcal{J}_T(D, B)$. Hence $\mathcal{J}_T(D, B) \rightarrow 0$ as $D \rightarrow 0$ for fixed B .

Proposition 3 (Certificate supplied by one local PIBE loss) Fix $k \in \{2, \dots, n\}$. Suppose Assumption 5 holds and

$$\mathcal{L}_D^{k,*} \leq \varepsilon_{D,k}, \quad \mathcal{L}_P^{k,*} \leq \varepsilon_{P,k}. \quad (97)$$

Define

$$\varepsilon_{D,k}^\# := \begin{cases} 2\varepsilon_{D,2} + 2\bar{w}^2, & k = 2, \\ \varepsilon_{D,k}, & k = 3, \dots, n, \end{cases} \quad (98)$$

and let L_{g_k} be a Lipschitz constant of e_{k-1}^{rec} . Set

$$\begin{aligned} D_{0,k} &:= \Psi_T(\varepsilon_{D,k}^\#, h_d, L_{g_k}), \\ D_{1,k} &:= \mathcal{J}_T(D_{0,k}, B_k), \\ R_k &:= \Psi_T(\varepsilon_{P,k}, h_r, L_{r_k}), \\ \eta_k &:= D_{1,k} + L_{f_{k-1}}^x D_{0,k} + R_k. \end{aligned} \quad (99)$$

Then

$$\|e_{k-1}^{\text{rec}}\|_\infty \leq D_{0,k}, \quad \|\dot{e}_{k-1}^{\text{rec}}\|_\infty \leq D_{1,k}, \quad \|r_k^*\|_\infty \leq R_k, \quad (100)$$

and the intrinsic new-state and parameter errors satisfy only the coupled certificate

$$\left\| \epsilon_{s_k}^{\text{int}} + \beta_k \epsilon_{p_{k-1}}^{\text{int}} \right\|_\infty \leq \eta_k. \quad (101)$$

If $\mathcal{L}_{Loc}^{k,*} = \mathcal{L}_D^{k,*} + \lambda \mathcal{L}_P^{k,*} \leq \varepsilon_k$, one may take $\varepsilon_{D,k} = \varepsilon_k$ and $\varepsilon_{P,k} = \varepsilon_k / \lambda$.

Proof 4 For $k \geq 3$, the samples in (92) are those of e_{k-1}^{rec} . For $k = 2$, the measurement equation and $(a+b)^2 \leq 2a^2 + 2b^2$ give

$$\frac{1}{N} \sum_{i=1}^N |e_1^{\text{rec}}(t_i)|^2 \leq 2\mathcal{L}_D^{2,*} + 2\bar{w}^2.$$

Lemma 2 therefore gives the first bound in (100); (96) gives the second, and another application of Lemma 2 to r_k^* gives the third.

Let $e_{s_k} = \bar{x}_k - x_k$ and $e_{p_{k-1}} = \bar{\theta}_{k-1} - \theta_{k-1}$. Subtracting the true $(k-1)$ -st equation of (1) from the exact-upstream residual identity and applying the mean-value formula in (89) gives the exact equality

$$e_{s_k}(t) + \beta_k(t)e_{p_{k-1}} = \dot{e}_{k-1}^{\text{rec}}(t) - a_k(t)e_{k-1}^{\text{rec}}(t) - r_k^*(t). \quad (102)$$

Using $|a_k| \leq L_{f_{k-1}}^x$ and (100) proves (101). $\square$

Remark 7 (A local cell does not identify the state-parameter split) No separate bound on $\epsilon_{s_k}^{\text{int}}$ or $\epsilon_{p_{k-1}}^{\text{int}}$ follows from (101), even under Assumption 6. Indeed, at the function-space level, consider the noise-free exact-upstream setting and choose a nonzero constant c small enough that $\theta_{k-1} + c$ and the outputs defined below remain admissible. Set

$$\bar{x}_{k-1}^k = x_{k-1}, \quad \bar{\theta}_{k-1} = \theta_{k-1} + c,$$

and choose the second decoder output as

$$\begin{aligned} \bar{x}_k(t) &= \dot{x}_{k-1}(t) \\ &\quad - f_{k-1}(x_1(t), \dots, x_{k-1}(t), \\ &\quad \quad \theta_1, \dots, \theta_{k-2}, \theta_{k-1} + c). \end{aligned}$$

Both the data loss and the physics loss are then zero, while the parameter error equals c and the state error compensates for it. An expressive decoder can approximate these function-space outputs arbitrarily closely, so restricting the optimization to neural networks does not restore a coercive local estimate in general. Thus, the previous claim that the measurement anchor alone forces the new state and parameter errors to be small was incorrect: the local loss anchors the reconstructed coordinate x_{k-1} , not the free new output x_k .

Corollary 1 (Conditional parameter certificate) Under Assumption 6, suppose an independent argument provides $\|\epsilon_{s_k}^{\text{int}}\|_\infty \leq S_k$. Then

$$|\epsilon_{p_{k-1}}^{\text{int}}| \leq \frac{\eta_k + S_k}{\sqrt{\gamma_k}}, \quad (103)$$

$$\|\epsilon_{s_k}^{\text{int}}\|_\infty \leq \eta_k + L_{f_{k-1}}^\theta |\epsilon_{p_{k-1}}^{\text{int}}|. \quad (104)$$

Proof 5 Let $q_k := \epsilon_{s_k}^{\text{int}} + \beta_k \epsilon_{p_{k-1}}^{\text{int}}$. With the normalized norm $\|v\|_{L_T^2} := (T^{-1} \int_0^T |v|^2 dt)^{1/2}$, Assumption 6, the triangle inequality, and (101) give

$$\sqrt{\gamma_k} |\epsilon_{p_{k-1}}^{\text{int}}| \leq \|\beta_k \epsilon_{p_{k-1}}^{\text{int}}\|_{L_T^2} = \|q_k - \epsilon_{s_k}^{\text{int}}\|_{L_T^2} \leq \eta_k + S_k.$$

This proves (103). Equation (104) follows directly from (101) and $|\beta_k| \leq L_{f_{k-1}}^\theta$. $\square$

Remark 8 (Behavior under weak excitation) The factor $1/\sqrt{\gamma_k}$ in the conditional bound (103) correctly diverges as $\gamma_k \rightarrow 0$. Thus, a positive lower bound on γ_k is required by this conditional parameter certificate, but Remark 7 shows that it is not sufficient to separate a free state output from the parameter using one local loss.

Remark 9 (Mesh-consistent loss interpretation) An empirical mean bound does not directly imply an N -independent $O(\sqrt{\varepsilon})$ supremum bound. The valid passage is Lemma 2, which uses both quasi-uniform grid coverage and temporal regularity. After optimizing over ρ , its generic small-error rate is of order $\varepsilon^{1/3}$ when the Lipschitz constant is fixed. This is weaker than the previously claimed square-root rate but is mathematically justified.

Remark 10 (Required global coercivity) To verify Assumption 4 rather than only (101), one needs a coercivity inequality for the stacked data-physics operator, or independent information on each new state. Assumption 2 supplies local conditioning for exact model trajectories, but applying it to PINN outputs with nonzero physics defects additionally requires a defect-to-solution stability estimate. Such a global argument, rather than the scalar local excitation condition alone, is the correct mechanism for separating the intermediate free state-parameter pairs. The final disturbance coefficient is different because no additional free state follows it; once the upstream errors are controlled, it admits the conditional weak-form certificate derived below.

4.3 Joint error propagation bound

For $k = 1, \dots, n$, define the normalized discrepancy between the recursively generated and exact-upstream inputs by

$$D_k^\ell := \left\| \check{\mathbf{U}}_k^\ell - \mathbf{U}_k^{\star, \ell} \right\|_{\mathcal{U}_k}. \quad (105)$$

Because the first cell receives the same measured trajectory in both evaluations, $D_1^\ell = 0$, even when the measurement is noisy. For $k = 2, \dots, n+1$, define the propagated and total full-cell errors

$$\begin{aligned} P_k^\ell &:= \left\| \check{\mathbf{Z}}_k^\ell - \bar{\mathbf{Z}}_k^\ell \right\|_{\mathcal{Z}_k}, \\ E_k^\ell &:= \left\| \check{\mathbf{Z}}_k^\ell - \mathbf{Z}_k^{\star, \ell} \right\|_{\mathcal{Z}_k}. \end{aligned} \quad (106)$$

Proposition 4 (Propagation through the trajectory-level bank) *Suppose Assumption 4 and the cell Lipschitz estimates (76) hold. Then, for every admissible trajectory ℓ and every $m = 2, \dots, n+1$,*

$$D_{m-1}^\ell \leq \sum_{k=2}^{m-1} \bar{\varepsilon}_k^{\text{int}} \prod_{j=k+1}^{m-1} (1 + \Lambda_j), \quad (107)$$

and

$$P_m^\ell \leq \Lambda_m \sum_{k=2}^{m-1} \bar{\varepsilon}_k^{\text{int}} \prod_{j=k+1}^{m-1} (1 + \Lambda_j), \quad (108)$$

$$E_m^\ell \leq \bar{\varepsilon}_m^{\text{int}} + \Lambda_m \sum_{k=2}^{m-1} \bar{\varepsilon}_k^{\text{int}} \prod_{j=k+1}^{m-1} (1 + \Lambda_j). \quad (109)$$

Empty sums are zero and empty products are one. In particular, $P_2^\ell = 0$: the first cell has no propagated input error, although its total error E_2^ℓ may be nonzero.

Proof 6 The cell Lipschitz estimate and the triangle inequality give

$$\begin{aligned} P_k^\ell &\leq \Lambda_k D_{k-1}^\ell, \\ E_k^\ell &\leq \bar{\varepsilon}_k^{\text{int}} + \Lambda_k D_{k-1}^\ell. \end{aligned} \quad (110)$$

For $k = 2, \dots, n$, the definition of the trajectory input yields the exact identity

$$\begin{aligned} (D_k^\ell)^2 &= (D_{k-1}^\ell)^2 + \frac{\|\check{\mathbf{X}}_k^{k, \ell} - \mathbf{X}_k^\ell\|_{N, 2}^2}{s_{x_k}^2} \\ &\quad + \frac{|\check{\theta}_{k-1}^{k, \ell} - \theta_{k-1}|^2}{s_{\theta_{k-1}}^2}. \end{aligned} \quad (111)$$

The square root of the last two terms is at most E_k^ℓ . Indeed, the normalized squared norm of a continuous trajectory is bounded by its supremum norm, and the new-state and parameter components are coordinate projections of the full-cell output. Therefore,

$$\begin{aligned} D_k^\ell &\leq D_{k-1}^\ell + E_k^\ell \\ &\leq (1 + \Lambda_k) D_{k-1}^\ell + \bar{\varepsilon}_k^{\text{int}}. \end{aligned} \quad (112)$$

Starting from $D_1^\ell = 0$, induction in (112) gives (107). Substitution into (110) gives (108) and (109). $\square$

For compactness, define the right-hand side of (109) by

$$\mathcal{B}_m := \bar{\varepsilon}_m^{\text{int}} + \Lambda_m \sum_{k=2}^{m-1} \bar{\varepsilon}_k^{\text{int}} \prod_{j=k+1}^{m-1} (1 + \Lambda_j). \quad (113)$$

Corollary 2 (Componentwise total-error bounds)

For $m = 2, \dots, n$, the recursively generated reported estimates satisfy

$$\|\check{x}_m^\ell - x_m^\ell\|_\infty \leq s_{x_m} \mathcal{B}_m, \quad (114)$$

$$|\check{\theta}_{m-1}^\ell - \theta_{m-1}| \leq s_{\theta_{m-1}} \mathcal{B}_m. \quad (115)$$

For the final cell,

$$\|\check{x}_n^{n+1, \ell} - x_n^\ell\|_\infty \leq s_{x_n} \mathcal{B}_{n+1}, \quad (116)$$

and

$$\|\check{a}^\ell - a^\ell\|_2 \leq s_a \mathcal{B}_{n+1}, \quad (117)$$

and hence

$$\|\check{d}_q^\ell - d^\ell\|_\infty \leq \varepsilon_{d, q} + G_q s_a \mathcal{B}_{n+1}. \quad (118)$$

Proof 7 The reported-state, parameter, auxiliary final-reconstruction, and coefficient inequalities follow by applying the corresponding coordinate projections of the scaled output norms (67)–(69) to (109). Finally,

$$\check{d}_q - d = \Gamma_q^\top (\check{a} - a) - r_q,$$

so (118) follows from $\|\Gamma_q^\top (\check{a} - a)\|_\infty \leq G_q \|\check{a} - a\|_2$ and $\|r_q\|_\infty \leq \varepsilon_{d, q}$. $\square$

4.4 Weak-form verification of the final coefficient

The final cell differs from the intermediate cells because its coefficient vector is not paired with a free subsequent state. We now derive a certificate directly for the recursively evaluated PIBE loss. Since f_n is locally Lipschitz on the compact admissible set, choose finite componentwise constants $L_{n, j}^x$ and $L_{n, j}^\theta$ such that

$$\begin{aligned} |f_n(\tilde{x}, \tilde{\theta}) - f_n(x, \theta)| &\leq \sum_{j=1}^n L_{n, j}^x |\tilde{x}_j - x_j| \\ &\quad + \sum_{j=1}^{n-1} L_{n, j}^\theta |\tilde{\theta}_j - \theta_j| \end{aligned} \quad (119)$$

for all admissible arguments.

Let

$$\varpi(t) := \frac{t(T-t)}{T^2}, \quad \psi_q(t) := \varpi(t) \Gamma_q(t). \quad (120)$$

Then $\psi_q(0) = \psi_q(T) = 0$. Define

$$\begin{aligned} W_{\Gamma, \varpi} &:= \int_0^T \varpi(t) \Gamma_q(t) \Gamma_q(t)^\top dt, \\ \gamma_{d, \varpi} &:= \lambda_{\min}(W_{\Gamma, \varpi}), \\ C_{\psi, 0} &:= \|\psi_q\|_{L^2(0, T; \mathbb{R}^q)}, \\ C_{\psi, 1} &:= \|\dot{\psi}_q\|_{L^2(0, T; \mathbb{R}^q)}. \end{aligned} \quad (121)$$

Because $\varpi > 0$ on $(0, T)$ and the components of Γ_q are linearly independent, $\gamma_{d,\varpi} > 0$. Thus, (121) introduces a conditioning constant but no additional identifiability assumption beyond (3).

For the recursively evaluated final cell, define the consistency residual

$$c_{n+1}(t) := \check{x}_n^{n+1}(t) - \check{x}_n^n(t) \quad (122)$$

and set

$$\begin{aligned} \check{x}^{n+1}(t) &:= \text{col}\left(\check{x}_1^2(t), \{\check{x}_j^j(t)\}_{j=2}^{n-1}, \check{x}_n^{n+1}(t)\right), \\ \check{\theta} &:= \text{col}\left(\{\check{\theta}_j^{j+1}\}_{j=1}^{n-1}\right). \end{aligned} \quad (123)$$

The middle state sequence is empty when $n = 2$. Let $\check{r}_{n+1}$ denote (40) evaluated at the recursive state, parameter, and coefficient outputs. The corresponding quadratic loss components are

$$\begin{aligned} \mathcal{L}_D^{n+1,\text{rec}} &:= \frac{1}{N} \sum_{i=1}^N |c_{n+1}(t_i)|^2, \\ \mathcal{L}_P^{n+1,\text{rec}} &:= \frac{1}{N_r} \sum_{j=1}^{N_r} |\check{r}_{n+1}(\tau_j)|^2. \end{aligned} \quad (124)$$

These are precisely the trajectory-wise PIBE terms in (38) and (39). Under Assumption 5, both c_{n+1} and $\check{r}_{n+1}$ are Lipschitz on $[0, T]$; denote valid constants by $L_{c,n+1}$ and $L_{\check{r},n+1}$, respectively. In particular, one may take $L_{c,n+1} \leq 2M_1$; finiteness of $L_{\check{r},n+1}$ follows from the M_1, M_2 bounds, local Lipschitz continuity of f_n on the compact argument set, $\Gamma_q \in C^1$, and compactness of $\mathcal{A}$.

Proposition 5 (Weak-form certificate for the final PIBE cell) *Suppose Assumption 5, the cell Lipschitz estimates (76), and the intrinsic accuracy bounds (80) for cells $2, \dots, n$ hold. Suppose also*

$$\mathcal{L}_D^{n+1,\text{rec}} \leq \varepsilon_{D,n+1}, \quad \mathcal{L}_P^{n+1,\text{rec}} \leq \varepsilon_{P,n+1}. \quad (125)$$

Let $C_{\text{mesh},d}$ and $C_{\text{mesh},r}$ be the mesh constants in Lemma 2 for the data and collocation grids, and define

$$\begin{aligned} A_{D,n+1} &:= \sqrt{TC_{\text{mesh},d}(\varepsilon_{D,n+1} + L_{c,n+1}^2 h_d^2)}, \\ A_{P,n+1} &:= \sqrt{TC_{\text{mesh},r}(\varepsilon_{P,n+1} + L_{\check{r},n+1}^2 h_r^2)}, \\ H_{n+1} &:= A_{D,n+1} + \sqrt{T} s_{x_n} \mathcal{B}_n. \end{aligned} \quad (126)$$

For $j = 1, \dots, n-1$, set $\iota(j) := \max\{2, j\}$, and define

$$\begin{aligned} F_{n+1} &:= L_{n,n}^x A_{D,n+1} \\ &+ \sqrt{T} \left(\sum_{j=1}^{n-1} L_{n,j}^x s_{x_j} \mathcal{B}_{\iota(j)} + L_{n,n}^x s_{x_n} \mathcal{B}_n \right. \\ &\quad \left. + \sum_{j=1}^{n-1} L_{n,j}^\theta s_{\theta_j} \mathcal{B}_{j+1} \right). \end{aligned} \quad (127)$$

Then the recursively generated coefficient satisfies

$$\begin{aligned} \|\check{a} - a\|_2 &\leq \frac{1}{\gamma_{d,\varpi}} \left[C_{\psi,1} H_{n+1} \right. \\ &\quad \left. + C_{\psi,0} (F_{n+1} + \sqrt{T} \varepsilon_{d,q} + A_{P,n+1}) \right]. \end{aligned} \quad (128)$$

Consequently,

$$\|\check{d}_q - d\|_\infty \leq \varepsilon_{d,q} + G_q C_{a,n+1}, \quad (129)$$

where $C_{a,n+1}$ denotes the right-hand side of (128). If $\mathcal{L}_D^{n+1,\text{rec}} + \lambda \mathcal{L}_P^{n+1,\text{rec}} \leq \varepsilon_{n+1}$, one may take $\varepsilon_{D,n+1} = \varepsilon_{n+1}$ and $\varepsilon_{P,n+1} = \varepsilon_{n+1}/\lambda$.

Proof 8 The L^2 part of Lemma 2 gives

$$\|c_{n+1}\|_{L^2(0,T)} \leq A_{D,n+1}, \quad \|\check{r}_{n+1}\|_{L^2(0,T)} \leq A_{P,n+1}. \quad (130)$$

Let

$$e_n^F := \check{x}_n^{n+1} - x_n.$$

Since

$$e_n^F = c_{n+1} + (\check{x}_n^n - x_n),$$

Corollary 2 and (130) give

$$\|e_n^F\|_{L^2(0,T)} \leq H_{n+1}. \quad (131)$$

Let

$$\Delta f_n(t) := f_n(\check{x}^{n+1}(t), \check{\theta}) - f_n(x(t), \theta),$$

where the recursive arguments are those used in $\check{r}_{n+1}$. The full-cell bound for cell 2 controls both $\check{x}_1^2 - x_1$ and $\check{x}_2^2 - x_2$; the remaining state and parameter arguments are controlled by Corollary 2. Applying (119), using $\|v\|_{L^2} \leq \sqrt{T}\|v\|_\infty$, and treating the current final-cell state through (131) yields

$$\|\Delta f_n\|_{L^2(0,T)} \leq F_{n+1}. \quad (132)$$

Subtracting the true last equation of (1) from the recursive final-cell residual gives, almost everywhere,

$$\check{r}_{n+1} = \dot{e}_n^F - \Delta f_n - \Gamma_q^\top (\check{a} - a) + r_q. \quad (133)$$

Multiply (133) by $\psi_q = \varpi \Gamma_q$, integrate over $[0, T]$, and use $\psi_q(0) = \psi_q(T) = 0$. Since $e_n^F \in W^{1,\infty}(0, T)$ and $\psi_q \in C^1([0, T])$, integration by parts gives

$$\begin{aligned} W_{\Gamma,\varpi}(\check{a} - a) &= - \int_0^T \dot{\psi}_q e_n^F dt - \int_0^T \psi_q \Delta f_n dt \\ &\quad + \int_0^T \psi_q r_q dt - \int_0^T \psi_q \check{r}_{n+1} dt. \end{aligned} \quad (134)$$

Cauchy-Schwarz, $\|r_q\|_{L^2} \leq \sqrt{T} \varepsilon_{d,q}$, and $\|W_{\Gamma,\varpi} v\|_2 \geq \gamma_{d,\varpi} \|v\|_2$, together with (130), (131), and (132), prove (128). The disturbance bound follows from $\check{d}_q - d = \Gamma_q^\top (\check{a} - a) - r_q$. $\square$

Remark 11 (Scope of the final-cell certificate)

Proposition 5 applies directly to the quadratic PIBE final-cell loss and to the recursively generated quantities used at inference. It requires upstream bounds $\mathcal{B}_2, \dots, \mathcal{B}_n$, but

it does not assume a second derivative of x_n or an intrinsic coefficient bound for the final cell. In particular, the coefficient bound converges to zero when the upstream bounds, the two sampled losses, the fill distances, and $\varepsilon_{d,q}$ converge to zero while $\gamma_{d,\varpi}$ remains bounded away from zero. For EPIBE, the same deterministic conclusion requires a separate condition converting the energy-based consistency loss into an L^2 bound on c_{n+1} .

Remark 12 (Growth with the chain length)

Suppose $\bar{\varepsilon}_k^{\text{int}} \leq \bar{\varepsilon}$ and $\Lambda_k \leq \Lambda$ for all cells. For $m \geq 3$ and $\Lambda > 0$, (107) gives

$$D_{m-1}^\ell \leq \bar{\varepsilon} \frac{(1+\Lambda)^{m-2} - 1}{\Lambda}, \quad P_m^\ell \leq \bar{\varepsilon}((1+\Lambda)^{m-2} - 1).$$

Thus, the worst-case estimate is geometric in the chain length. Lipschitz-constrained training can reduce its base [26, 8], but the inequality does not justify the stronger claim that merely enforcing $\Lambda \leq 1$ prevents geometric amplification. Moreover, Assumption 4 is a post-training hypothesis; increasing network width or depth alone does not guarantee that its constants decrease.

Remark 13 (Disturbance estimation) The final cell is indexed by $n+1$. Equation (118) is the direct propagation bound, conditional on the intrinsic final-cell constant in Assumption 4. Proposition 5 provides the alternative loss-based bound (129); it replaces that final intrinsic coefficient constant by the achieved PIBE losses, the basis conditioning constant, and the upstream bounds $\mathcal{B}_2, \dots, \mathcal{B}_n$. Both bounds retain the irreducible remainder $\varepsilon_{d,q}$. Because their numerical sizes also depend on scales, basis conditioning, and cell Lipschitz constants, neither formulation implies that the final-cell bound must be larger than every intermediate bound.

5 Simulations and results

In this section, we detail the technical aspects of implementing the previously introduced algorithms. We begin with data generation, then illustrate the algorithms using a 6th-order nonlinear system. We then discuss the results and present a sensitivity analysis with respect to dataset size and the number of training trajectories, as we believe these are important aspects of data-driven algorithms for control and observer design.

5.1 Data generation

For training the model, we generate system trajectories from arbitrary initial conditions sampled from an initial-condition subset $\chi_0 \subseteq \chi \subset R^n$, according to Assumption 1. We then generate the trajectories on a finite time horizon $T > 0$ by numerically solving the nonlinear dynamical system described in Eq. 1. The data generation follows the steps described below :

- (1) First, we define a set $\chi_0 \subseteq \chi$, where χ_0 is the initial conditions subset, from which we arbitrarily choose initial conditions $x_0 = [x_1^0, \dots, x_n^0]^T$.
- (2) Simulate system 1 for each initial condition, and from $t_0 = 0$ to $t_F = T$ to obtain the corresponding trajectories and output :

$$\begin{cases} x^i(t) := x(t_k, x_0^i, \theta) \\ y^i(t) := x_1^i(t) \end{cases} \quad (135)$$

for $k = 0, \dots, F$ and $i = 1, \dots, P$, where P is the number of trajectories generated and F the number of time steps.

- (3) Split the generated trajectories and corresponding measurements (outputs) into training and validation data, where $\Omega^{\text{train}} \cap \Omega^{\text{test}} = \emptyset$, Ω describing the data.

In the numerical study, the known basis vector Γ_q introduced in (2) is chosen as a B-spline basis. Let

$$\Xi = \{\xi_0, \xi_1, \dots, \xi_{q+m}\}$$

be a nondecreasing knot vector on $[0, T]$, and let $m \geq 2$ denote the spline degree. The interior-knot multiplicities are selected so that the resulting basis functions are continuously differentiable, consistently with the regularity imposed in (2).

Definition 1 For $i = 0, \dots, q+m-1$, define

$$\gamma_{i,0}(t) = \begin{cases} 1, & \xi_i \leq t < \xi_{i+1}, \\ 0, & \text{otherwise,} \end{cases}$$

and, for $\ell = 1, \dots, m$ and $i = 0, \dots, q+m-\ell-1$, define

$$\begin{aligned} \gamma_{i,\ell}(t) = & \frac{t - \xi_i}{\xi_{i+\ell} - \xi_i} \gamma_{i,\ell-1}(t) \\ & + \frac{\xi_{i+\ell+1} - t}{\xi_{i+\ell+1} - \xi_{i+1}} \gamma_{i+1,\ell-1}(t), \end{aligned} \quad (136)$$

where a term is defined to be zero whenever its denominator is zero. The values at $t = T$ are defined by continuous extension from the left.

The q basis functions of degree m are collected as

$$\Gamma_q(t) = [\gamma_{0,m}(t) \cdots \gamma_{q-1,m}(t)]^\top, \quad (137)$$

so that the structured disturbance component is

$$d_q(t) = \Gamma_q(t)^\top a = \sum_{i=0}^{q-1} a_i \gamma_{i,m}(t), \quad (138)$$

and the complete disturbance remains $d = d_q + r_q$ as in (2). Decreasing the knot spacing or increasing q may reduce the approximation error $\varepsilon_{d,q}$, but the basis dimension and knot vector are fixed during estimation. Increasing q also enlarges the class of admissible disturbances and may reduce the distinguishability between parameter and disturbance effects; this trade-off is addressed through the identifiability conditions introduced subsequently. To illustrate the previously introduced algorithm, we consider the following system:

6 CONCLUSION

This paper introduces a chain-of-estimators framework for estimating states, parameters, and disturbances in a class of nonlinear dynamical systems. Two frameworks are proposed to address different noise scenarios. A PINN-based chain is used with noise-free or compactly supported truncated-Gaussian measurements, while the EPIBE extension combines each PINN cell with a constrained scalar EBM for flexible pooled residual-density fitting. In the reported simulations, PIBE performs well in the truncated-Gaussian and noise-free scenarios but loses accuracy in the tested non-Gaussian cases, whereas EPIBE yields smaller state and disturbance errors. These numerical findings do not constitute a general robustness or convergence guarantee. Future work will focus on proving statistical and optimization consistency of the joint estimator, extending the framework to online estimation, and improving its scaling properties.

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